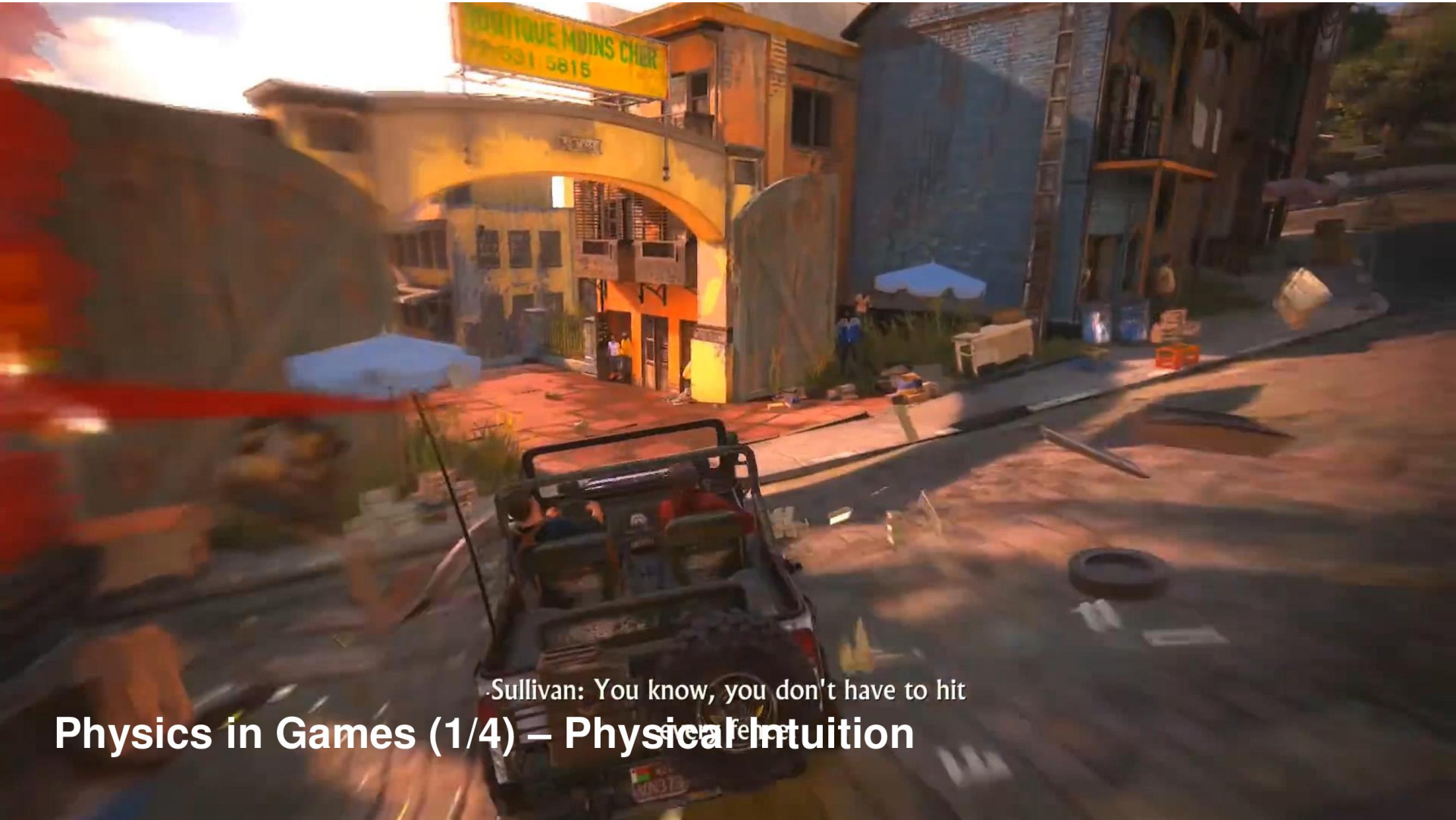




Physics System

Basic Concepts



Sullivan: You know, you don't have to hit

Physics in Games (1/4) – Physical Intuition

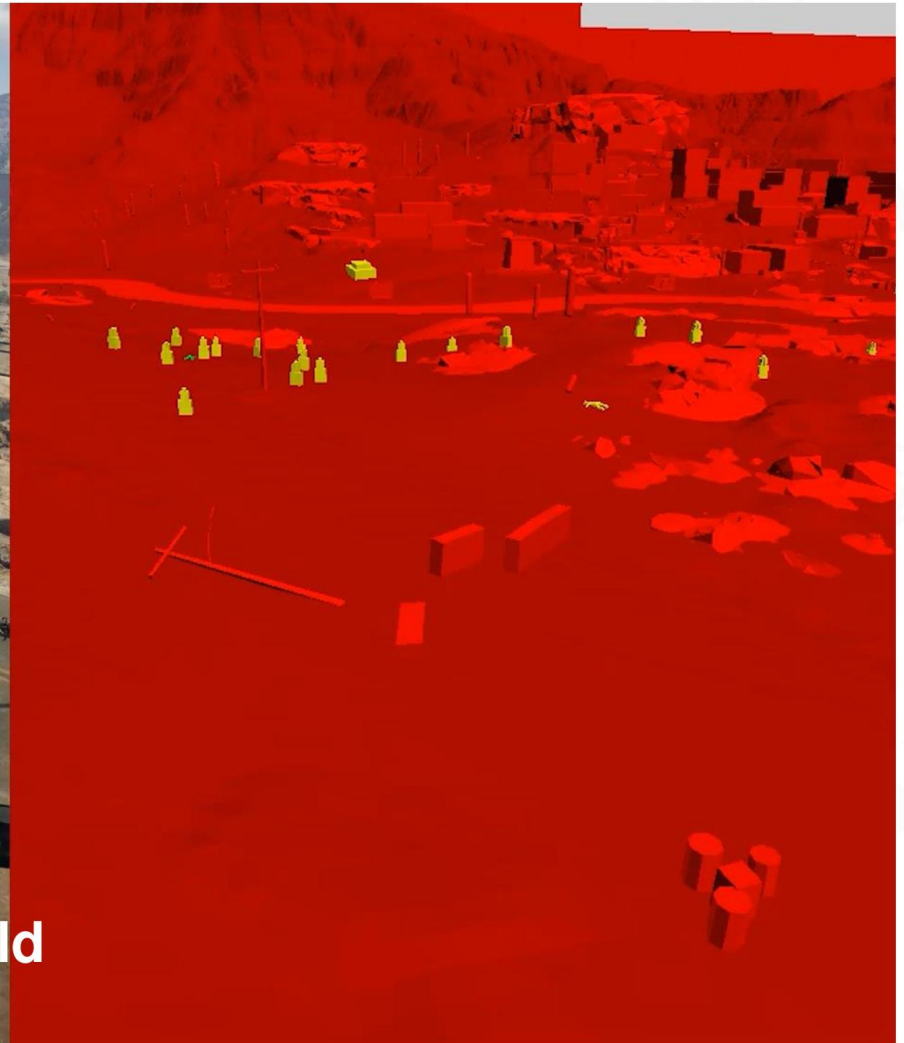
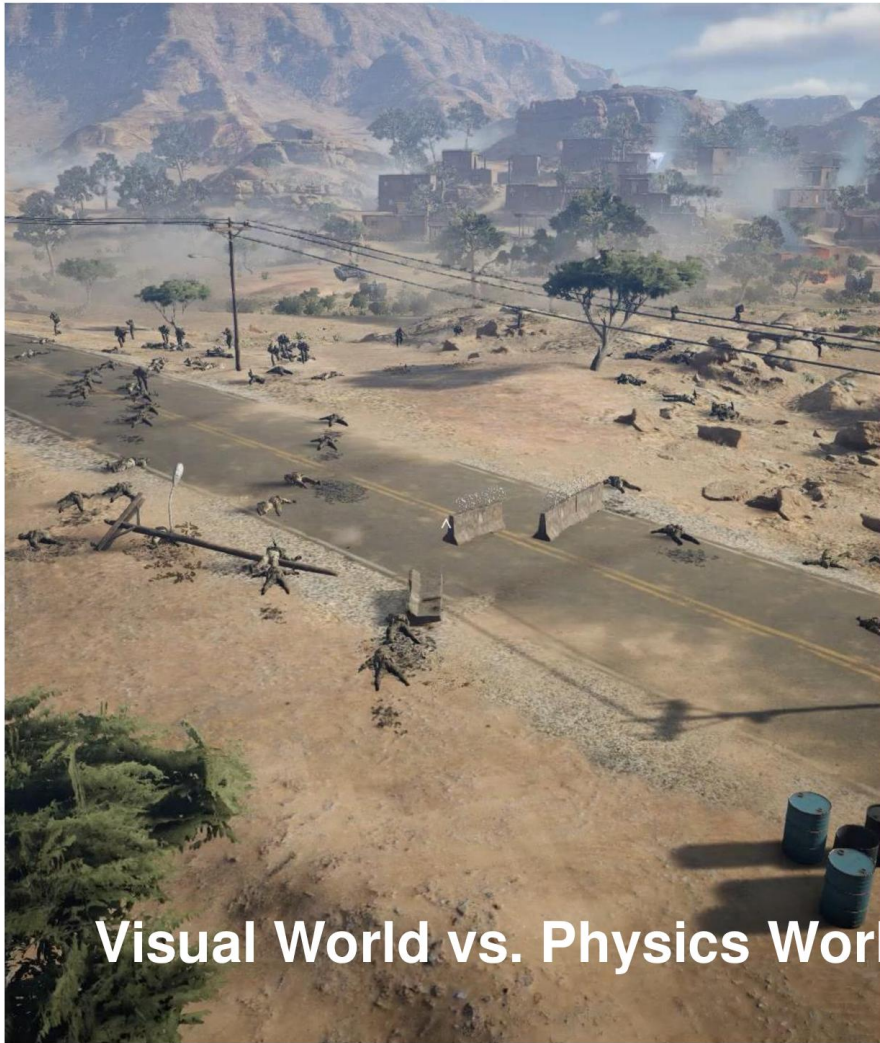
🗺️ ? Lieu à découvrir (250 m)

Physics in Games (4/4) – Artistic





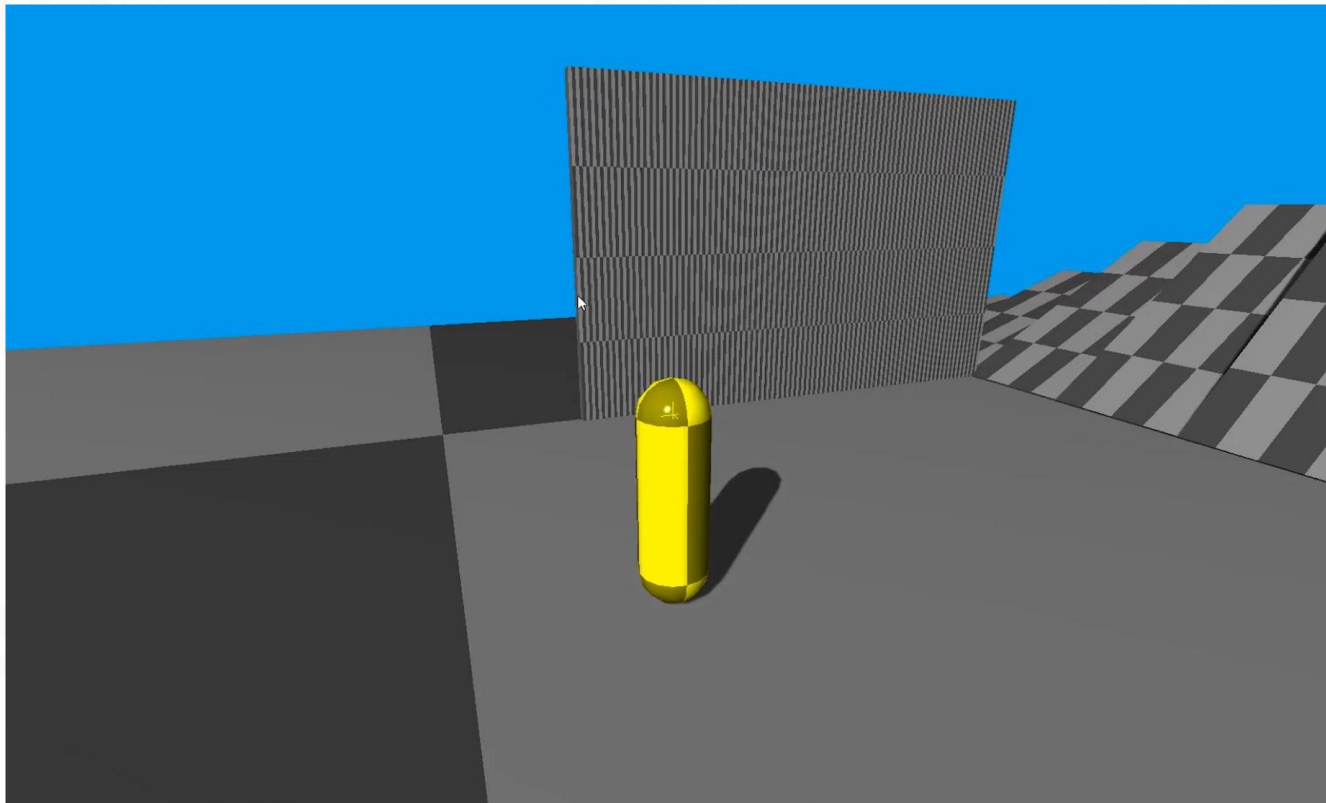
Physics Actors and Shapes



Visual World vs. Physics World

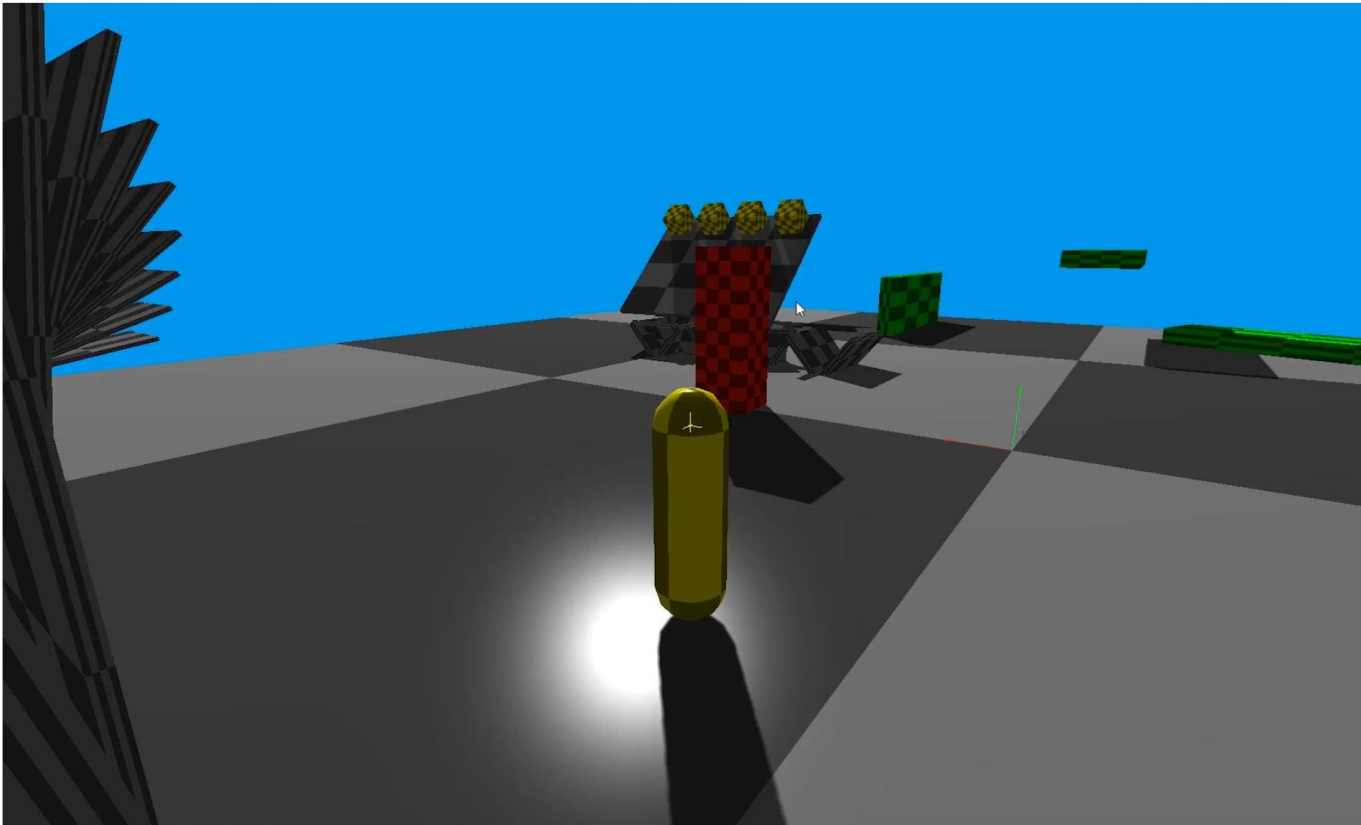


Actor – Static





Actor – Dynamic





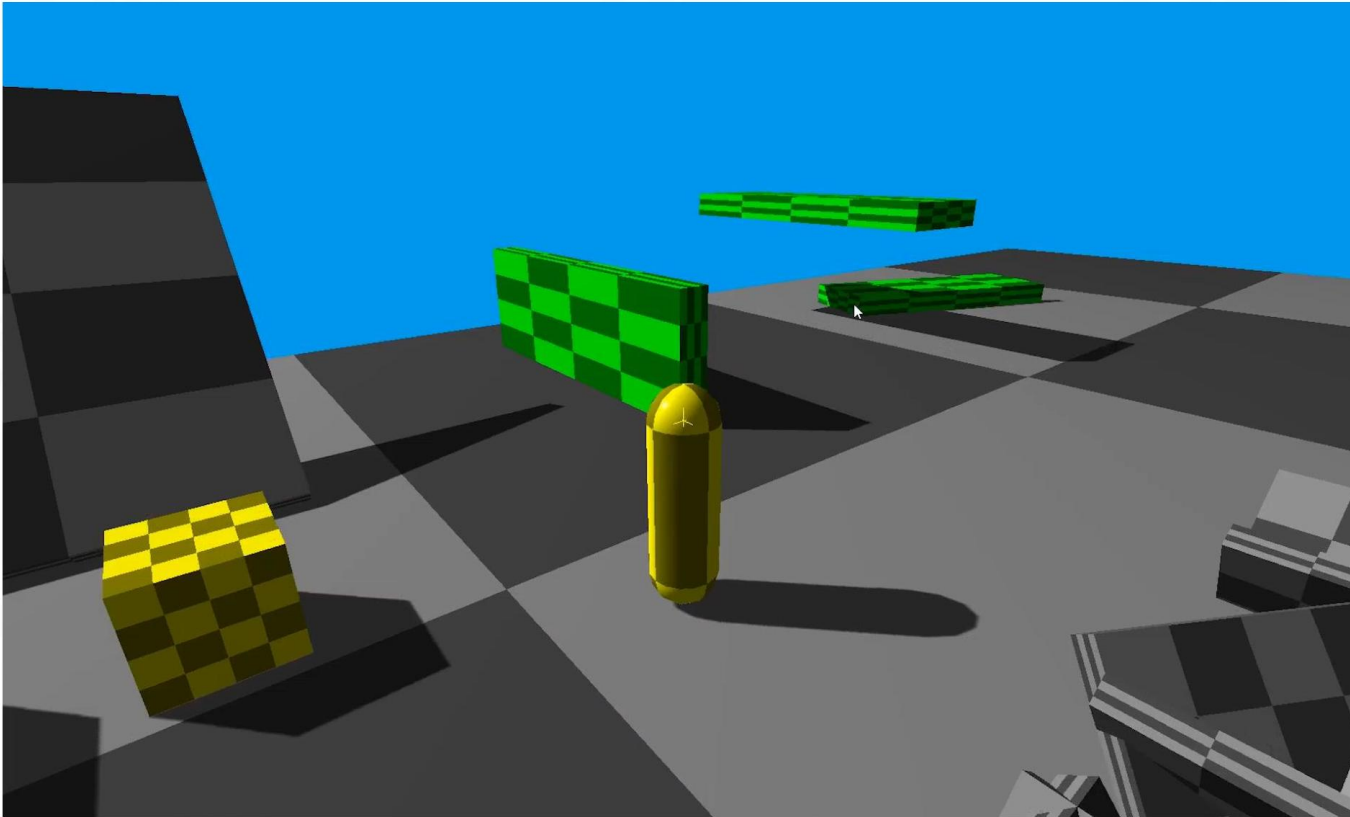
Trigger

- Like static actor, not moving
- But not blocking
- Notifies when actors enter or exit





Actor – Kinematic (No Physics Law)





Actor – Summary

Static Actor

- Not moving

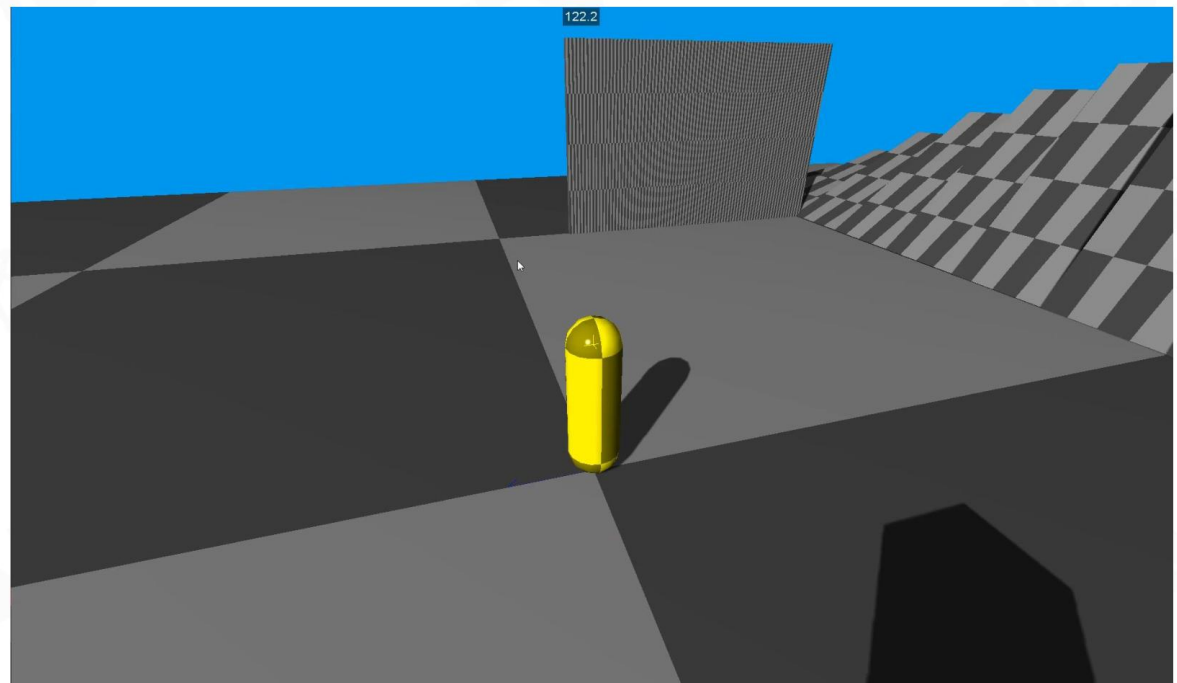
Dynamic Actor

- Can be affected by forces/torques/impulses

Trigger

Kinematic Actor

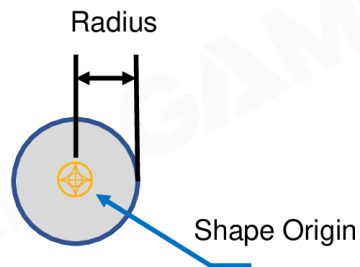
- Ignoring physics rules
- Controlled by gameplay logic directly



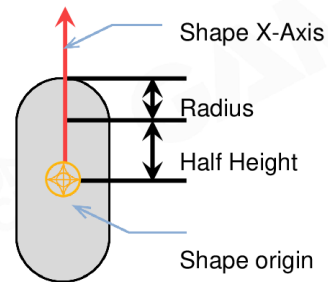


Actor Shapes

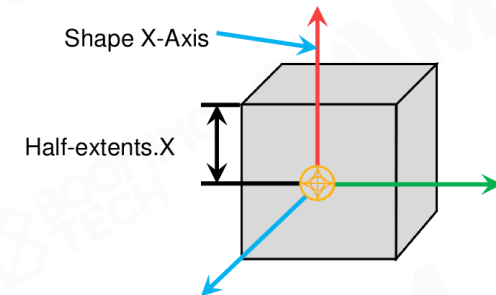
Spheres



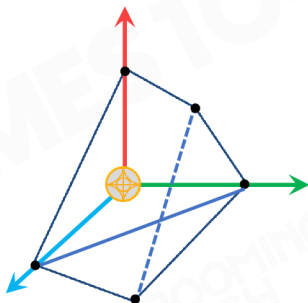
Capsules



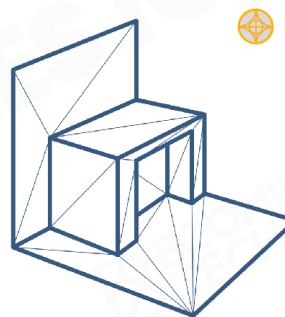
Boxes



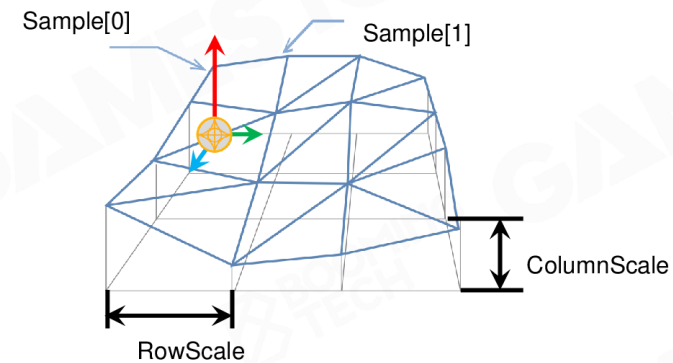
Convex Meshes



Triangle Meshes

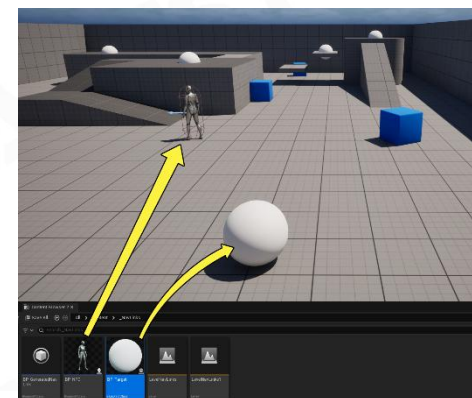
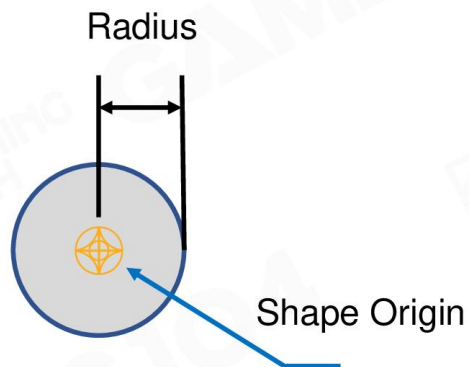


Height Fields



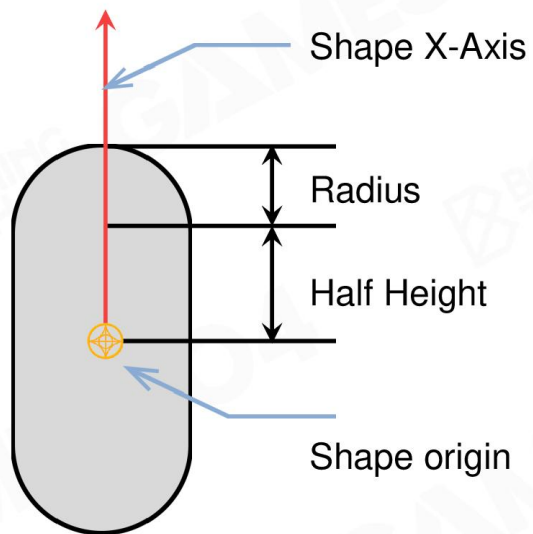


Shapes – Spheres



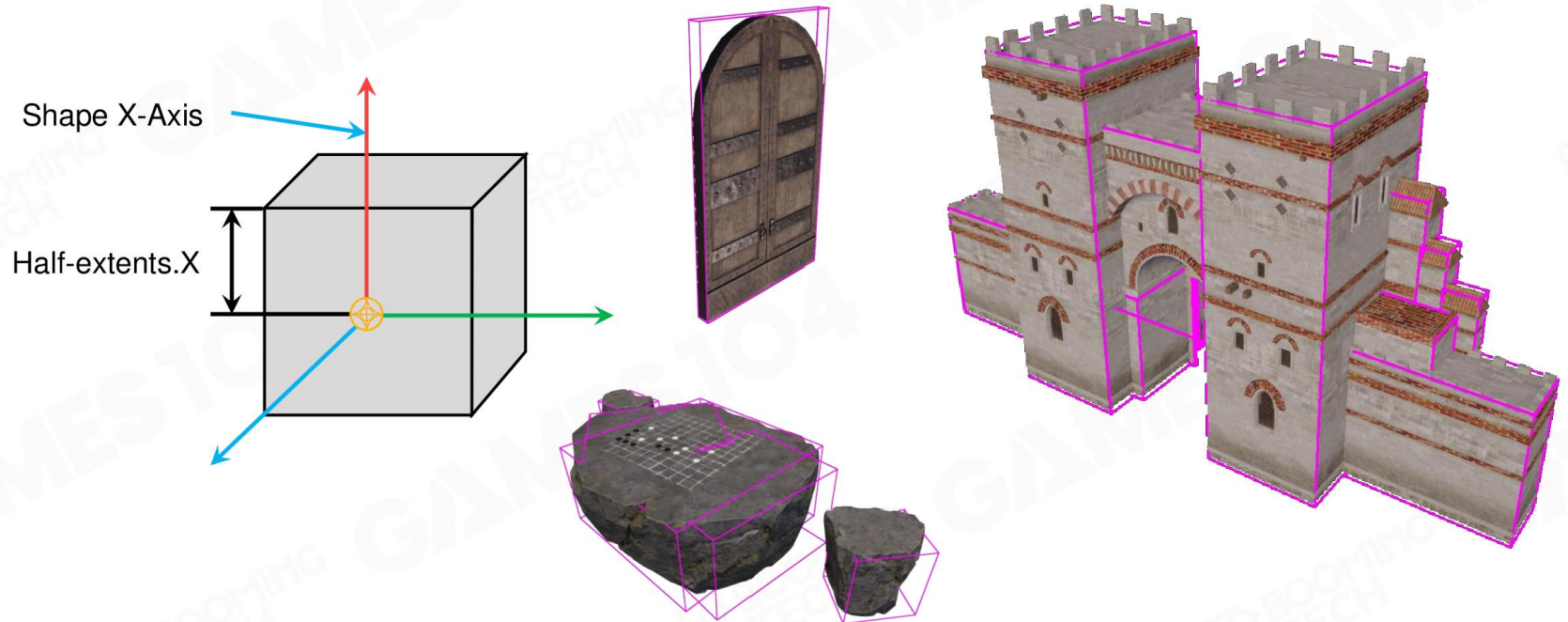


Shapes – Capsules



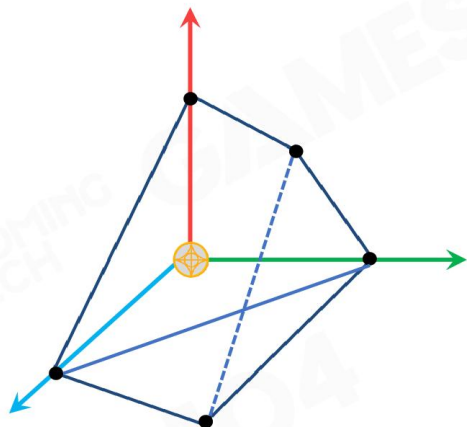


Shapes – Boxes





Shapes – Convex Meshes

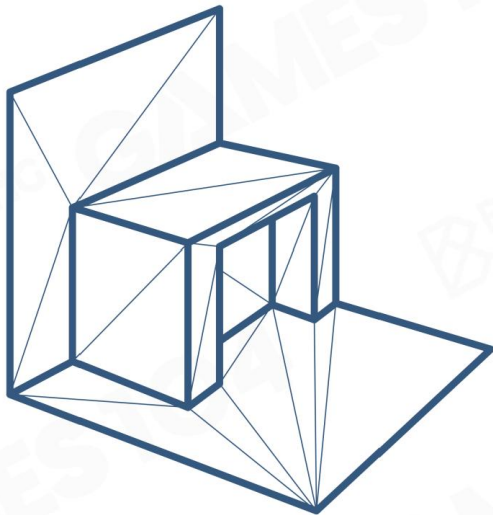


Vertices and faces limits of
convex meshes





Shapes – Triangle Meshes



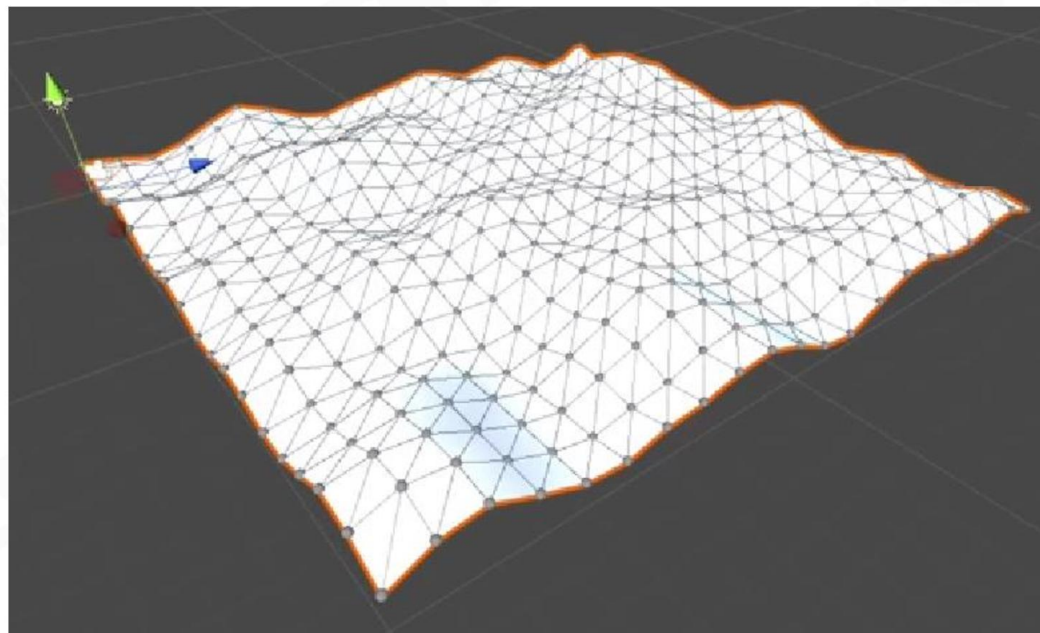
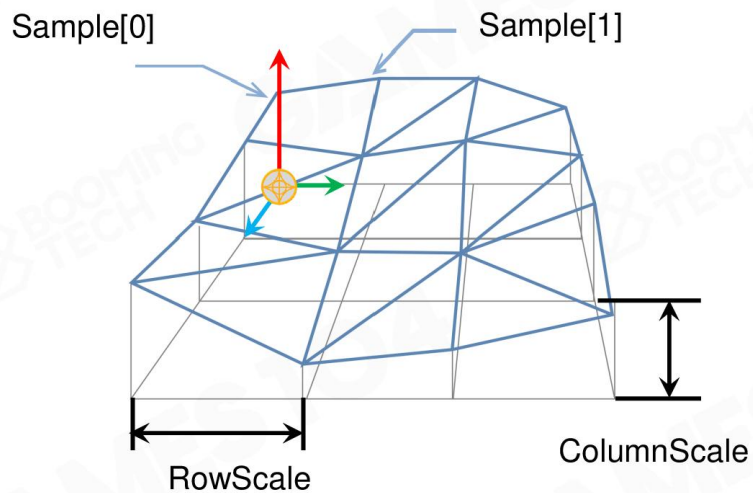
- Dynamic actors can't have triangle meshes

Physics engines (like NVIDIA PhysX, Havok, or Bullet) require convex shapes for dynamic bodies because:

- **Concave geometry causes ambiguous collisions:** when objects interpenetrate or collide along irregular concave surfaces, resolving forces and contacts becomes non-deterministic.
- **Triangle meshes lack volume:** they represent only surfaces, so the engine cannot easily compute physical properties such as mass, inertia, or internal volume.
- **Performance concerns:** continuous collision detection with arbitrary concave meshes would be extremely slow, as it would require testing each triangle against others every frame.



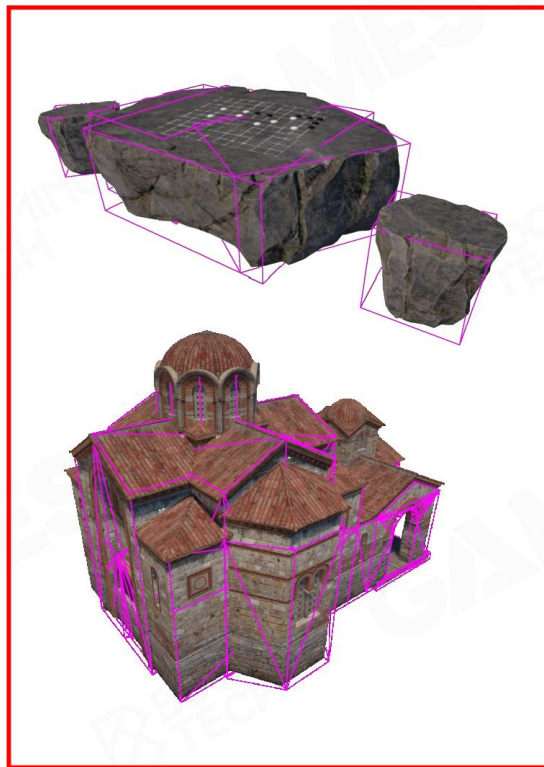
Shapes – Height Fields





Wrap Objects with Physics Shapes

- Approximated Wrapping
 - Don't need to be perfect
- Simplicity
 - Prefer simple shapes (avoid triangle mesh if possible)
 - Least shapes





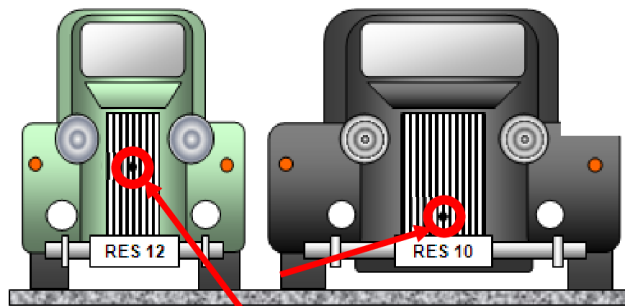
Shape Properties – Mass and Density



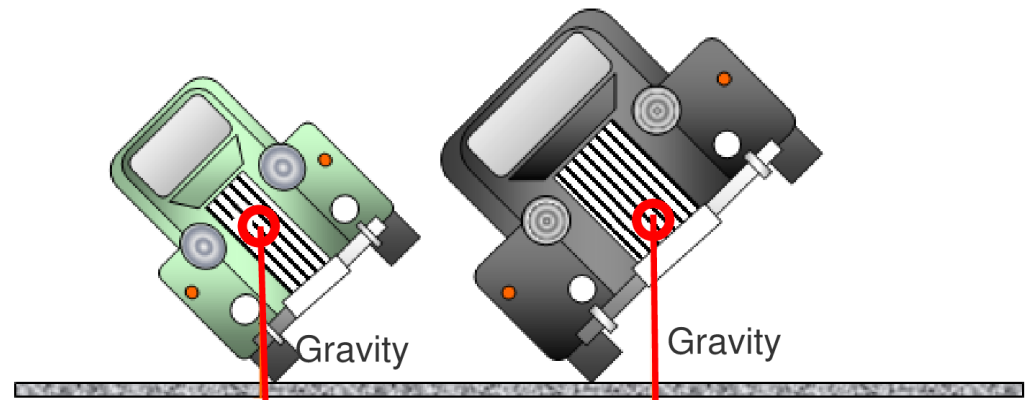
Gomboc Shape



Shape Properties - Center of Mass



Center of Mass

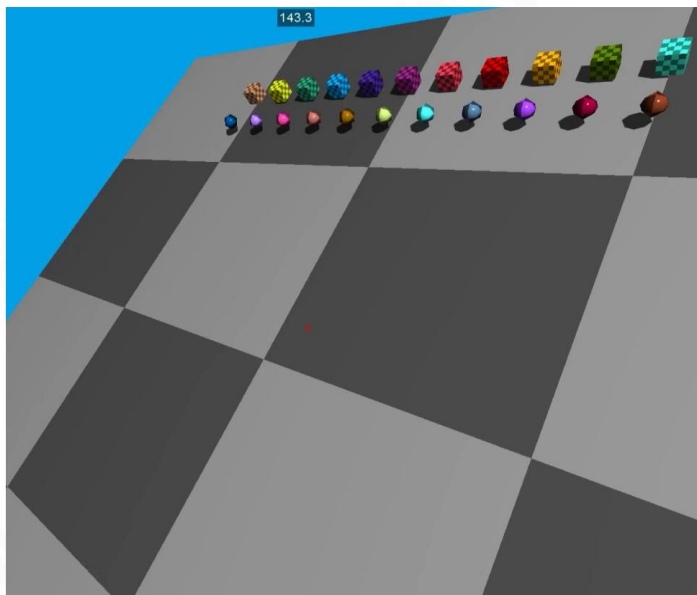


Topple

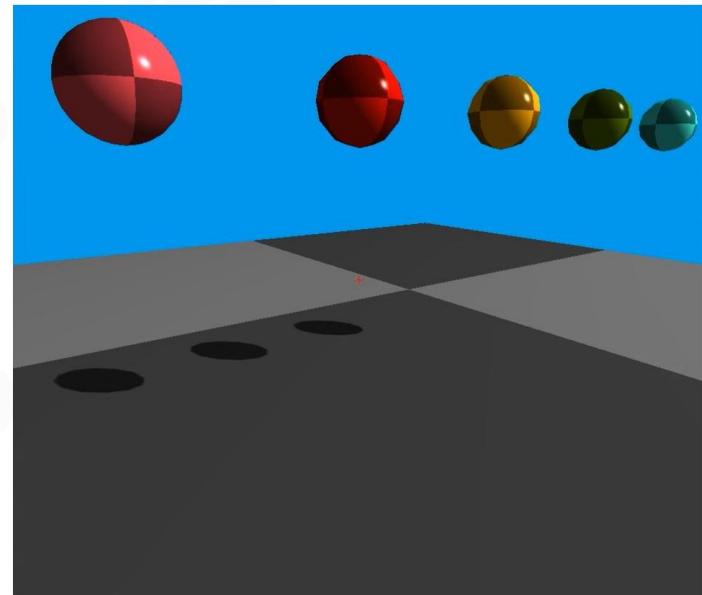
Not Topple



Shape Properties – Friction & Restitution



Different Friction Parameters



Different Restitution Parameters



Forces



Force

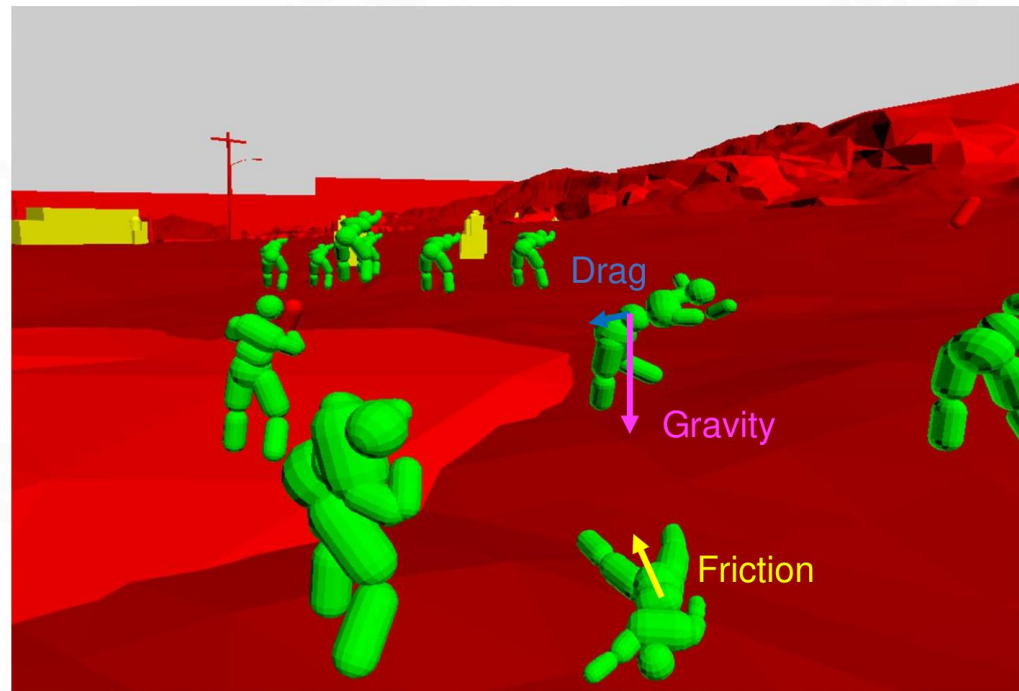
- We can apply forces to give dynamic objects accelerations, therefore affecting their movements
- Examples
 - Gravity
 - Drag
 - Friction
 - ...





Force

- We can apply forces to give dynamic objects accelerations, therefore affecting their movements
- Examples
 - Gravity
 - Drag
 - Friction
 - ...





Impulse

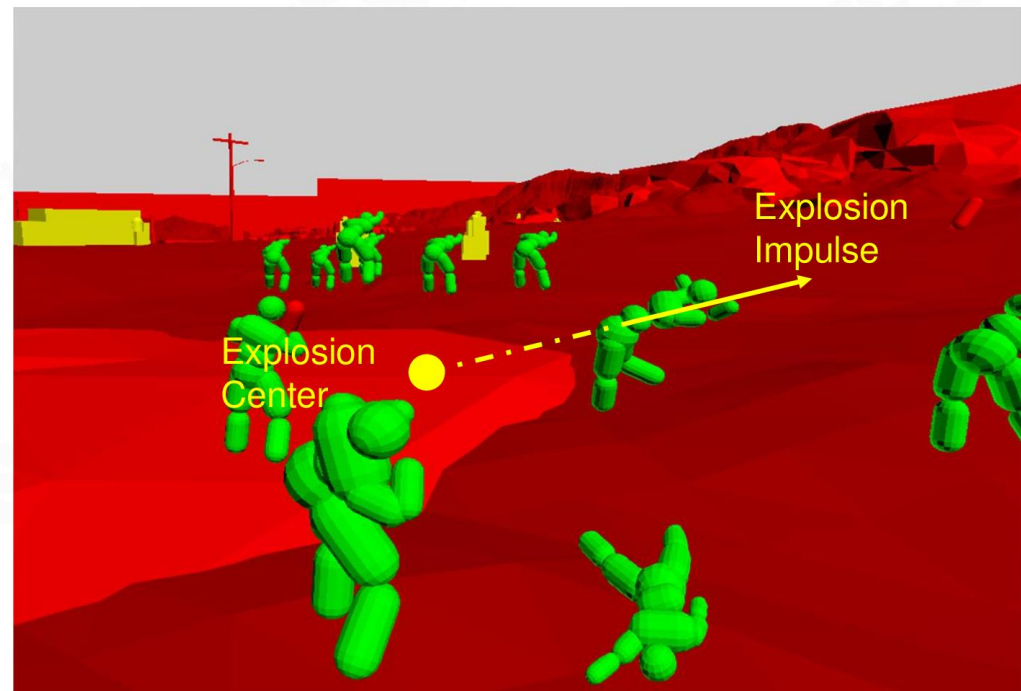
- We can change velocity of actors immediately by applying impulses
- E.g. simulating an explosion





Impulse

- We can change velocity of actors immediately by applying impulses
- E.g. simulating an explosion





Movements

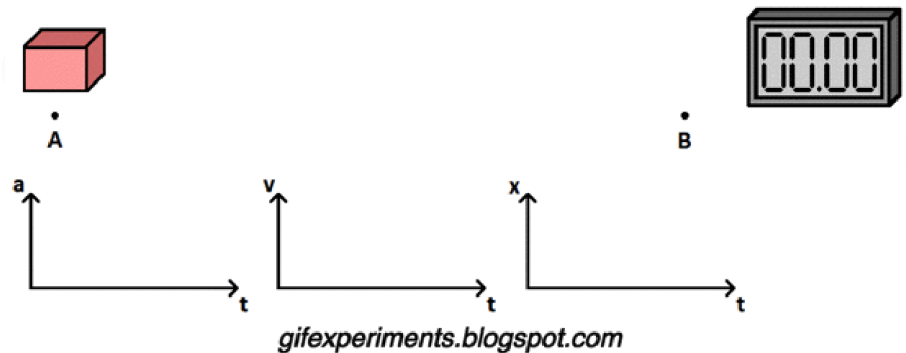


Newton's 1st Law of Motion

If there is no external force

$$\vec{v}(t + \Delta t) = \vec{v}(t)$$

$$\vec{x}(t + \Delta t) = \vec{x}(t) + \vec{v}(t)\Delta t$$





Newton's 2st Law of Motion

If there is external force

$$\vec{F} = m\vec{a}$$

Force Mass Acceleration

$$\vec{a}(t) = \frac{d\vec{v}(t)}{dt} = \frac{d^2\vec{x}(t)}{dt^2}$$



A



B



gifexperiments.blogspot.com



Movement under Constant Force

$$\vec{F} = m \vec{a}$$

$$\vec{a} = \vec{F} / m$$

$$\vec{v}(t + \Delta t) = \vec{v}(t) + \vec{a}(t)\Delta t$$

$$\vec{x}(t + \Delta t) = \vec{x}(t) + \vec{v}(t)\Delta t + \frac{1}{2}\vec{a}(t)\Delta t^2$$



Movement under Varying Force

Newton's 2st Law of Motion

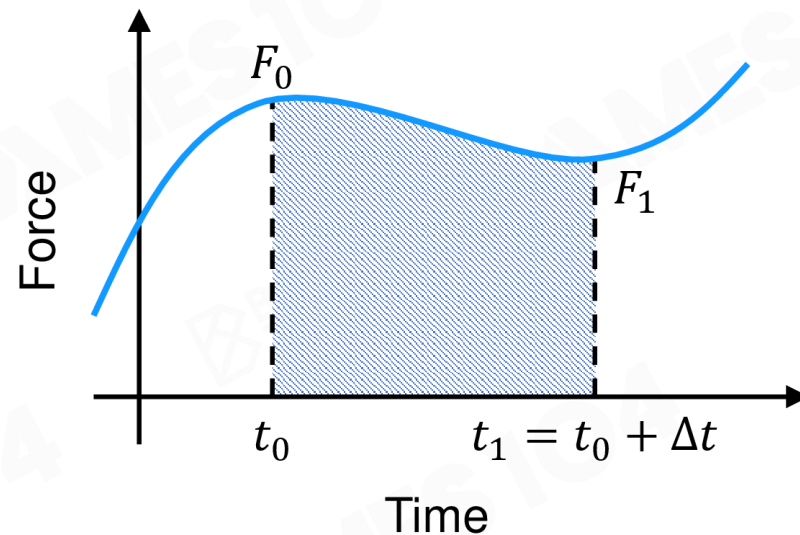
If there is *varying* external force

$$\vec{F} = m \vec{a}$$

$$\vec{a} = \vec{F}/m$$

$$\vec{v}(t + \Delta t) = \vec{v}(t) + \text{?}$$

$$\vec{x}(t + \Delta t) = \vec{x}(t) + \text{?}$$





Movement under Varying Force

Newton's 2st Law of Motion

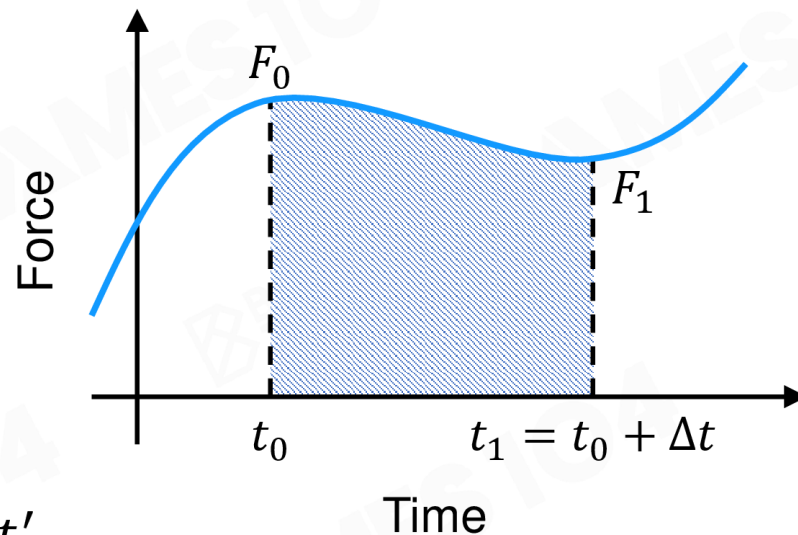
If there is *varying* external force

$$\vec{F} = m \vec{a}$$

$$\vec{a} = \vec{F}/m$$

$$\vec{v}(t + \Delta t) = \vec{v}(t) + \int_t^{t+\Delta t} \vec{a}(t') dt'$$

$$\vec{x}(t + \Delta t) = \vec{x}(t) + \int_t^{t+\Delta t} \vec{v}(t') dt'$$

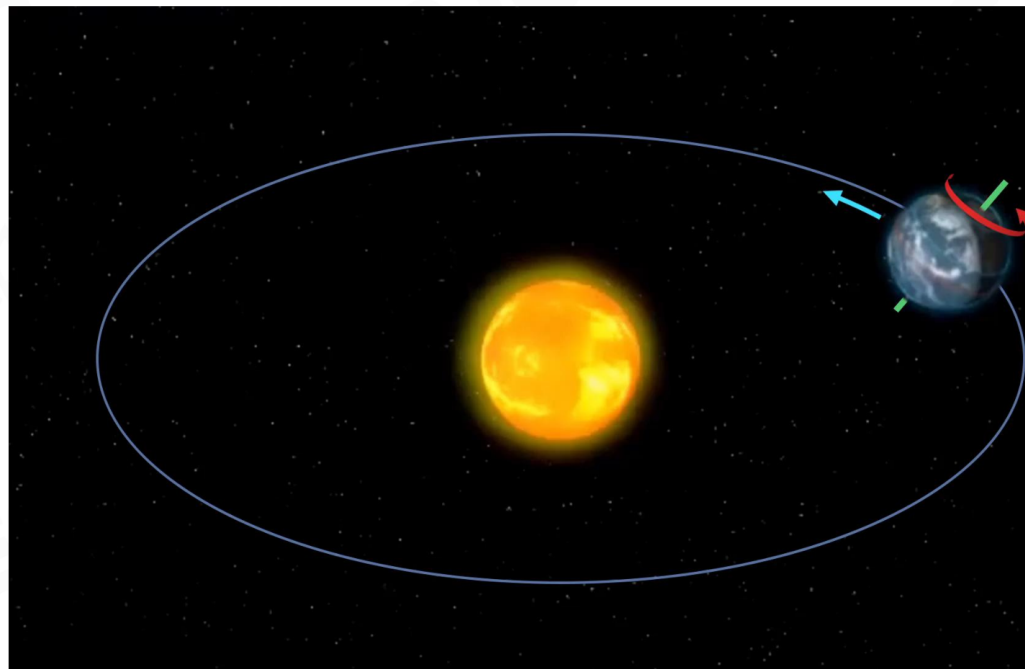




Example of Simple Movement

- Position
- Orientation
- Linear Velocity
- Angular Velocity

$$\mathbf{X}(t) = \begin{pmatrix} \vec{x}(t) \\ R(t) \\ \vec{v}(t) \\ \vec{\omega}(t) \end{pmatrix}$$



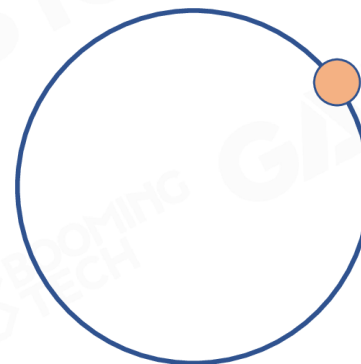
Earth In The Solar System



Motion in Reality

At time t

- Position: $\vec{x}(t)$
- Linear Velocity: $\vec{v}(t) = \frac{d\vec{x}(t)}{dt}$





Simulation in Game

At time t

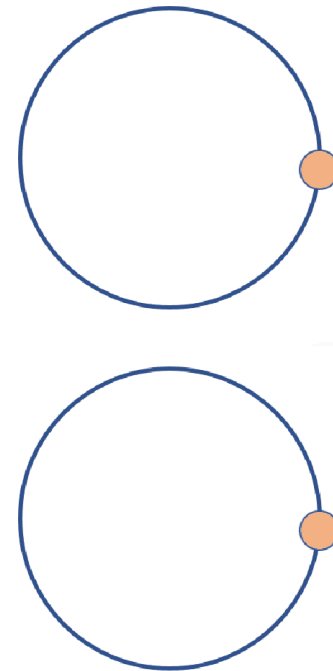
- Position: $\vec{x}(t)$
- Linear Velocity: $\vec{v}(t) = \frac{d\vec{x}(t)}{dt}$

Simulation Step

Given $\vec{x}(t), \vec{v}(t)$

Compute $\vec{x}(t + \Delta t), \vec{v}(t + \Delta t)$

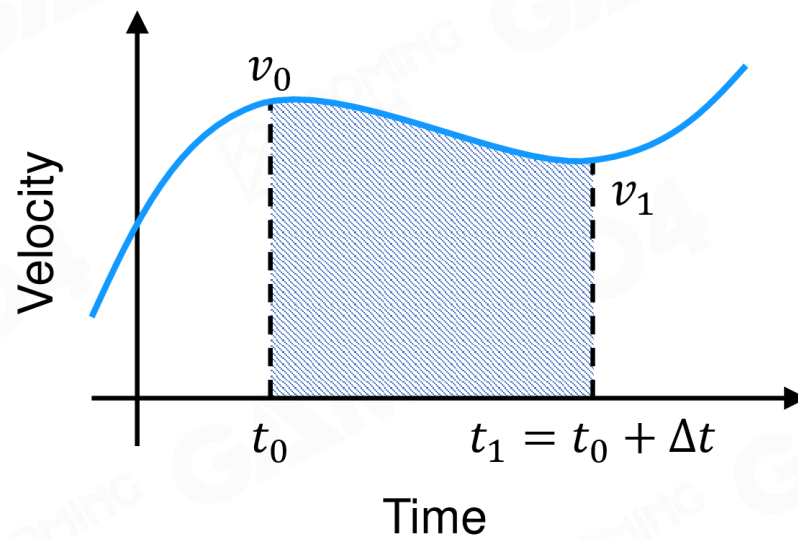
Δt is the time step size





Time Integration

$$\vec{x}(t_1) = \vec{x}(t_0) + \int_{t_0}^{t_1} \vec{v}(t) dt$$





Euler's Method

200

— 442 — (248 —

C A P V T VII.

METHODVS GENERALIS

INTEGRALIA QVAECVNQVE PROXIME

INVENIENDI.

Problema 36.

297.

Formulae integralis cuiuscunque $y = \int X dx$ valorem vero proxime indagare.

Solutio.

Cum omnis formula integralis per se sit indeterminata, ea semper ita determinari solet, ut si variabili x certus quidam valor puta a tribuatur, ipsam integrale $y = \int X dx$ datum valorem puta b obtineat. Integratione igitur hoc modo determinata, quaestio huc redit, si variabili x alius quicunque valor ab a diuersus tribuatur, valor, quem tum integrale y sit habiturum, definiatur. Tribuamus ergo ipsi x primo valorem parum ab a discrepantem, puta $x = a + a'$, ut a sit quantitas valde parva: et quia functio X parum variatur, siue pro x scribatur a siue $a + a'$ eam tanquam constantem spectare licet. Hinc ergo formulae differentialis $X dx$ integrale

C A P V T VII.

201

integrale erit $Xx + \text{Const.} = y$; sed quia posito $c = a$ fieri debet $y = b$, et valor ipsius X quasi nactus immutatus, erit $Xa + \text{Const.} = b$, ideoque $\text{Const.} = b - Xa$, unde consequimur $y = b + X(x - a)$. Quare si ipsi x valorem $a + a'$ tribuamus, habebimus valorem conuenientem ipsius y , qui sit $b + \beta$; c iam simili modo ex hoc casu definire poterimus y , si ipsi x tribuatur alius valor parum superans $a + a'$, posito igitur $a + a'$ loco x , valor ipsius X inde ortus denovo pro constante haberi poterit, indeque fiet $y = b + \beta + X(x - a - a')$. Hanc igitur operationem continuare licet quousque lubet, cuius ratio quo melius percipiat, rem ita praesentemus:

$$\begin{aligned} x = a \quad \text{fiat} \quad X = A \quad \text{et} \quad y = b \\ x = a' \quad \dots X = A' \quad \dots y = b' = b + A(a' - a) \\ x = a'' \quad \dots X = A'' \quad \dots y = b'' = b' + A'(a'' - a') \\ x = a''' \quad \dots X = A''' \quad \dots y = b''' = b'' + A''(a''' - a'') \\ \text{etc.} \end{aligned}$$

si valores a, a', a'', a''' etc. secundum differentias valde parvas procedere ponuntur. Erit ergo $y = b + A(a' - a)$ quippe in quam obit formula aucta $y = b + X(x - a)$ fit enim $X = A$, qui ponitur $x = a$, tum vero tribuitur ipsi x valor a' , cui respondet $y = b'$, simili modo erit $b'' = A'(a'' - a')$; tum $b''' = b'' + A''(a''' - a'')$ etc. vti supra posuimus.

C c

C A P V T VII.

testituendo ergo valores praecedentes habeo

$$\begin{aligned} A(a' - a) \\ A(a' - a) + A'(a'' - a') \\ A(a' - a) + A'(a'' - a') + A''(a''' - a'') \\ A(a' - a) + A'(a'' - a') + A''(a''' - a'') + A'''(a^{(4)} - a''') \\ \text{etc.} \end{aligned}$$

x quantumvis excedet a , series a', a'', a''' etc. continuetur ad x , et vltimum aggregatum lozem ipsius y .

Coroll. 1.

98. Si incrementa, quibus x augetur, statuatur scilicet a , ut sit $a' = a + a$, $a + a$, $a'' = a + 3a$, etc. quibus valoribus instituitur functio X abeat in A, A', A'' etc. limus illorum valorum puta $a + na$ sit x vero X , erit $b + a(A + A' + A'' + A''' + \dots + X)$.

Coroll. 2.

99. Valor ergo integralis y per summationem seriei $A, A', A'' \dots X$, cuius termini X formantur ponendo loco x successus $a, a + a, a + 2a, \dots a + na$, eruitur. Summa enim illius seriei per differentiam a multiplicata et ad b adiecta dabit valorem ipsius y , qui ipsi $x = a + na$ respondet.

Coroll. 3.



Leonhard Euler

1707-1783

Institutiones calculi integralis (1768-70), p200-203.



Explicit (Forward) Euler's Method (1/3)

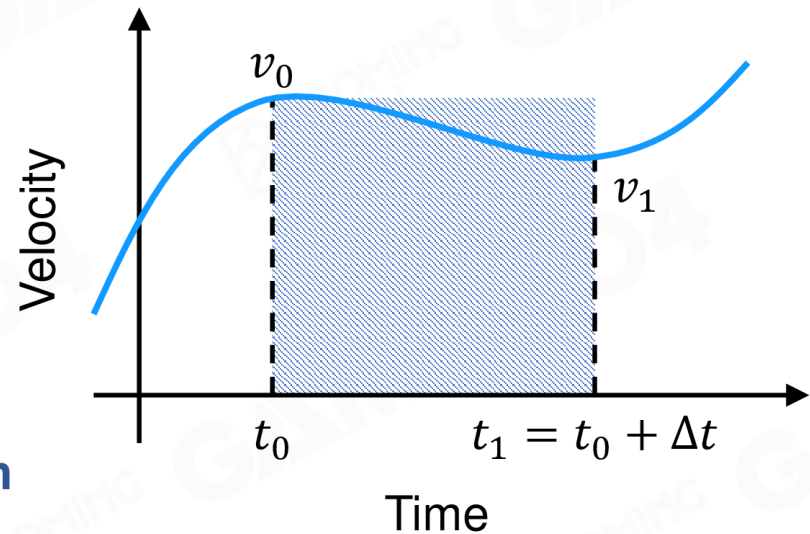
Simplest estimation

Assume the force is constant
during the time step

$$\begin{cases} \vec{v}(t_1) = \vec{v}(t_0) + M^{-1} \vec{F}(t_0) \Delta t \\ \vec{x}(t_1) = \vec{x}(t_0) + \vec{v}(t_0) \Delta t \end{cases}$$

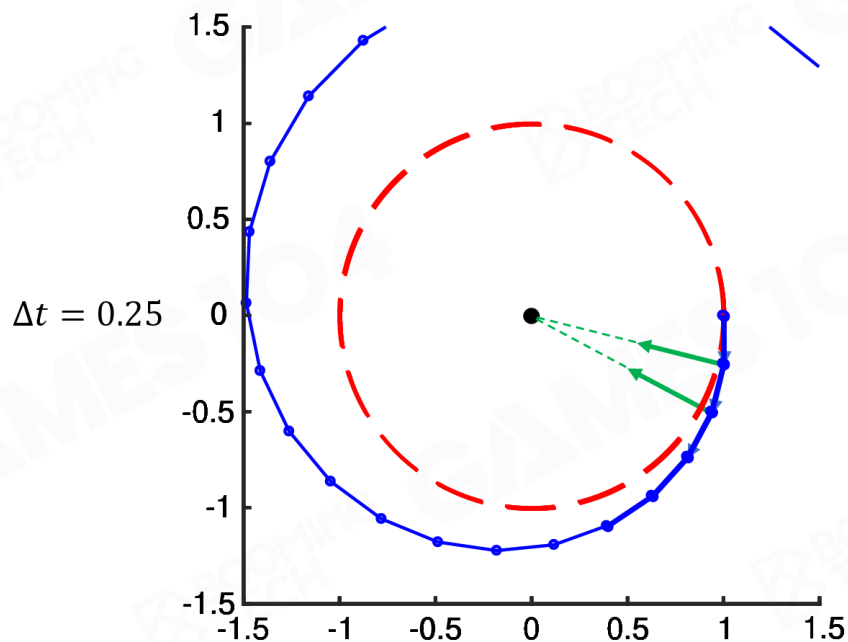
Current States

All quantities are known



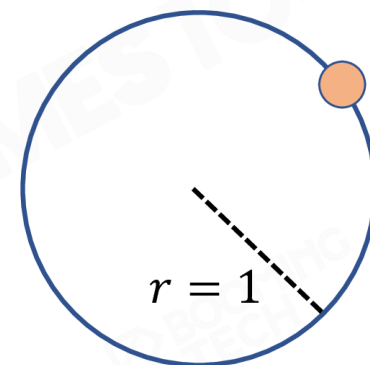


Explicit (Forward) Euler's Method (2/3)



Example:

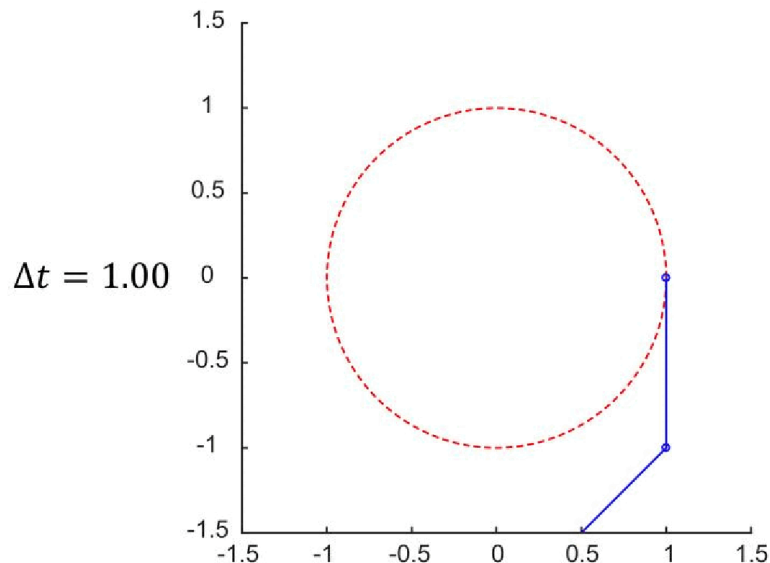
A particle moving around a circle





Explicit (Forward) Euler's Method (3/3)

The result of explicit Euler's method explodes!



Pros:

- Easy to calculate, efficient

Cons:

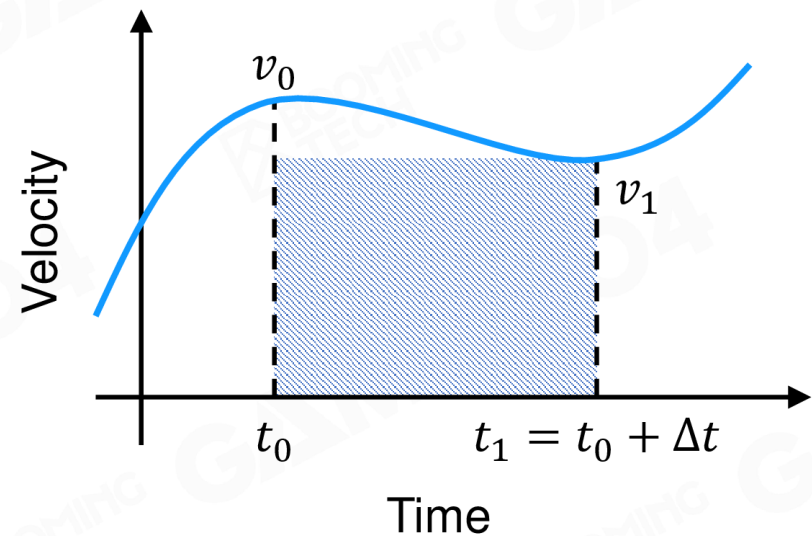
- Poor stability
- Energy growing as time progresses



Implicit (Backward) Euler's Method (1/2)

$$\begin{cases} \vec{v}(t_1) = \vec{v}(t_0) + M^{-1} \vec{F}(t_1) \Delta t \\ \vec{x}(t_1) = \vec{x}(t_0) + \vec{v}(t_1) \Delta t \end{cases}$$

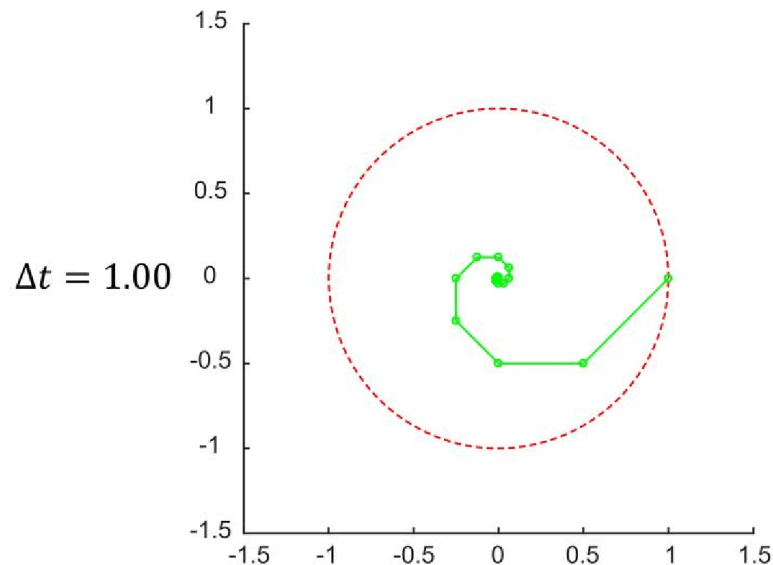
**Future states
Unknown yet**





Implicit (Backward) Euler's Method (2/2)

The result of implicit Euler's method spirals!



Pros:

- Unconditionally stable

Cons:

- Expensive to solve
- Challenging to implement when non-linearity presents
- Energy attenuates as time progresses



Semi-implicit Euler's Method (1/2)

Explicit Euler's Method

$$\begin{cases} \vec{v}(t_1) = \vec{v}(t_0) + M^{-1} \vec{F}(t_0) \Delta t \\ \vec{x}(t_1) = \vec{x}(t_0) + \vec{v}(t_0) \Delta t \end{cases}$$



Implicit Euler's Method

$$\begin{cases} \vec{v}(t_1) = \vec{v}(t_0) + M^{-1} \vec{F}(t_1) \Delta t \\ \vec{x}(t_1) = \vec{x}(t_0) + \vec{v}(t_1) \Delta t \end{cases}$$



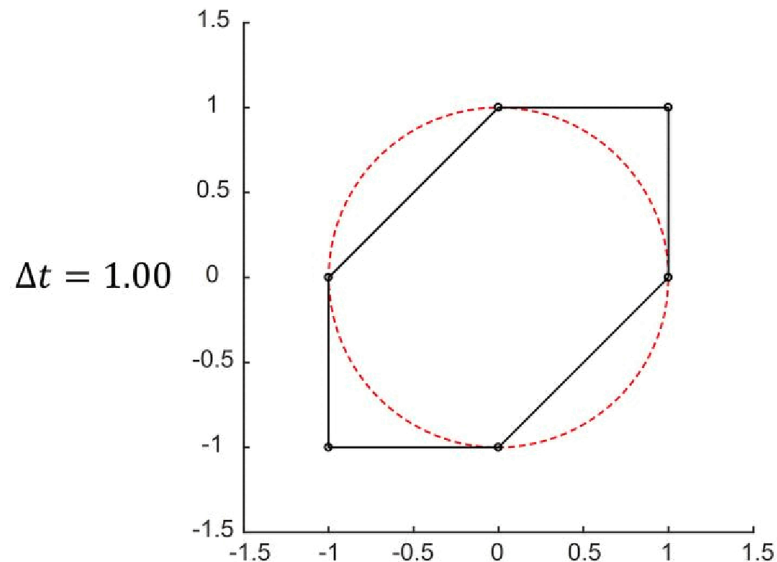
Current States

$$\begin{cases} \vec{v}(t_1) = \vec{v}(t_0) + M^{-1} \vec{F}(t_0) \Delta t \\ \vec{x}(t_1) = \vec{x}(t_0) + \vec{v}(t_1) \Delta t \end{cases}$$

Future states

Semi-implicit Euler's Method (2/2)

The result approximates the circle well if the timestep is small enough



- Conditionally stable
- Easy to calculate, efficient
- Preserves energy as time progresses



Rigid Body Dynamics



Particle Dynamics

- Position $\vec{x}$
- Linear Velocity $\vec{v} = \frac{d\vec{x}}{dt}$
- Acceleration $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{x}}{dt^2}$
- Mass M
- Momentum $\vec{p} = M\vec{v}$
- Force $\vec{F} = \frac{d\vec{p}}{dt} = M\vec{a}$





Rigid body Dynamics

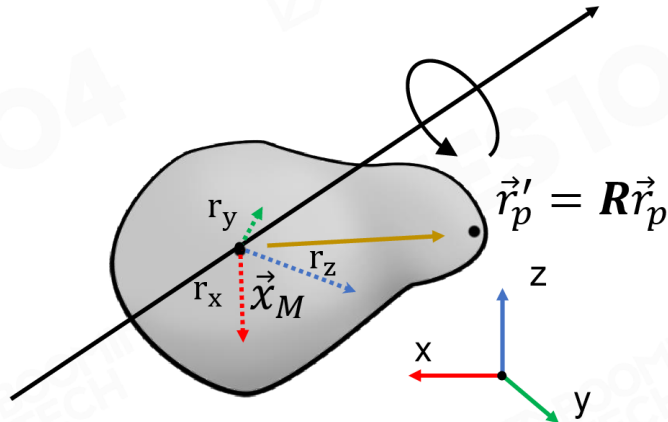
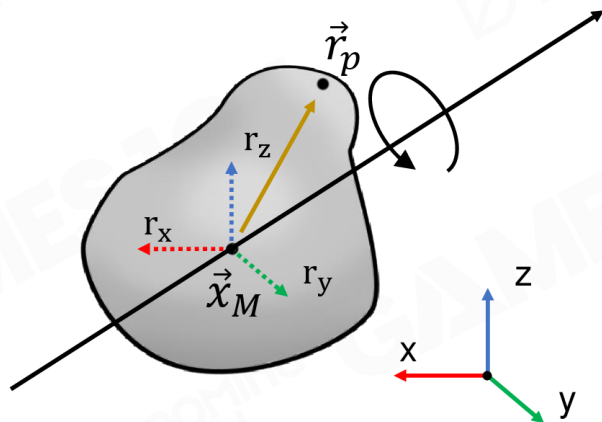
Besides linear values, rigid body dynamics have angular values

- Orientation $\mathbf{R}$
- Angular velocity $\vec{\omega}$
- Angular acceleration $\vec{\alpha}$
- Inertia tensor $\mathbf{I}$
- Angular momentum $\vec{L}$
- Torque $\vec{\tau}$



Orientation – R

A matrix $\mathbf{R}(t) = \begin{bmatrix} r_{xx} & r_{yx} & r_{zx} \\ r_{xy} & r_{yy} & r_{zy} \\ r_{xz} & r_{yz} & r_{zz} \end{bmatrix}$ or a quaternion $q = [s, \vec{v}]$





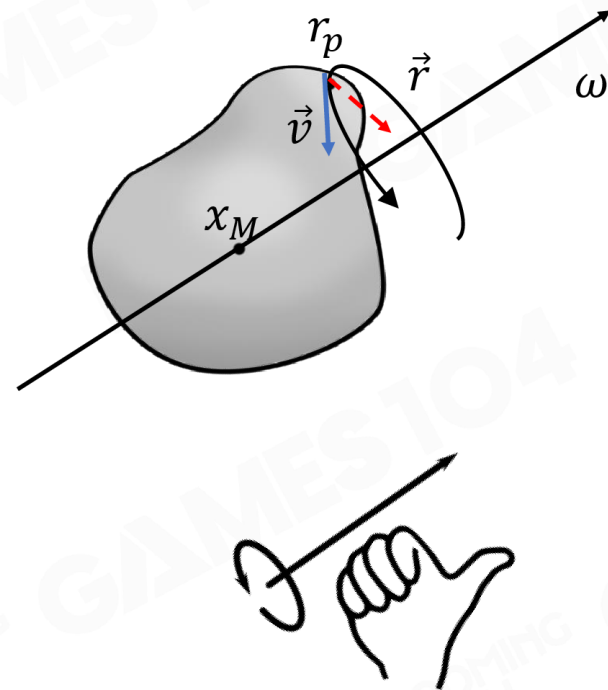
Angular Velocity – $\vec{\omega}$

Direction of $\vec{\omega}$ is the direction of the rotation axis

θ : rotated angle in radians

$$\|\vec{\omega}\| = \frac{d\theta}{dt}$$

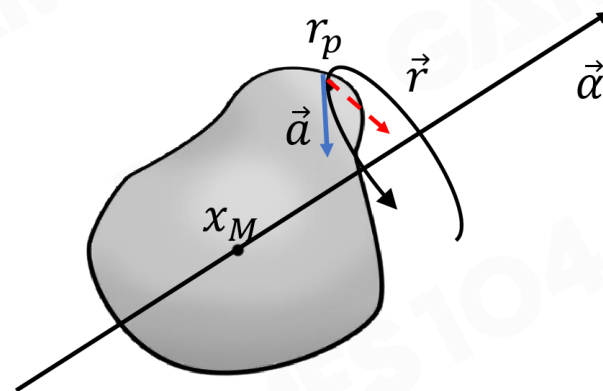
$$\vec{\omega} = \frac{\vec{v} \times \vec{r}}{\|\vec{r}\|^2}$$





Angular Acceleration – $\vec{\alpha}$

$$\vec{\alpha} = \frac{d\vec{\omega}}{dt} = \frac{\vec{a} \times \vec{r}}{\|\vec{r}\|^2}$$

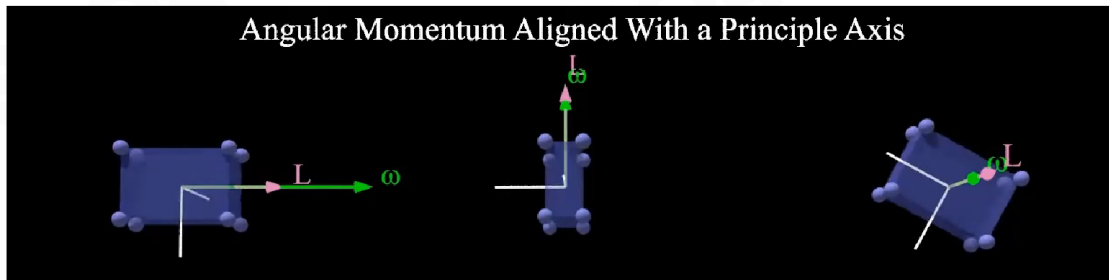




Rotational Inertia – I (1/2)

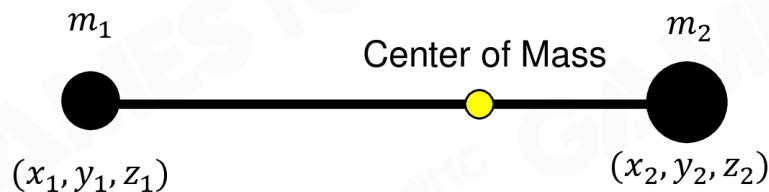
- Rotational inertia describes the distribution of mass for a rigid body

$$\mathbf{I} = \mathbf{R} \cdot \mathbf{I}_0 \cdot \mathbf{R}^T$$





Rotational Inertia – I (2/2)



Total Mass:

$$M = m_1 + m_2$$

Center of Mass:

$$CoM = \frac{m_1}{M} (x_1, y_1, z_1) + \frac{m_2}{M} (x_2, y_2, z_2)$$

Initial Inertia Tensor:

$$I_0 = \begin{bmatrix} m_1(y_1^2 + z_1^2) + m_2(y_2^2 + z_2^2) & -m_1x_1y_1 - m_2x_2y_2 & -m_1x_1z_1 - m_2x_2z_2 \\ -m_1y_1x_1 - m_2y_2x_2 & m_1(x_1^2 + z_1^2) + m_2(x_2^2 + z_2^2) & -m_1y_1z_1 - m_2y_2z_2 \\ -m_1z_1x_1 - m_2z_2x_2 & -m_1z_1y_1 - m_2z_2y_2 & m_1(x_1^2 + y_1^2) + m_2(x_2^2 + y_2^2) \end{bmatrix}$$



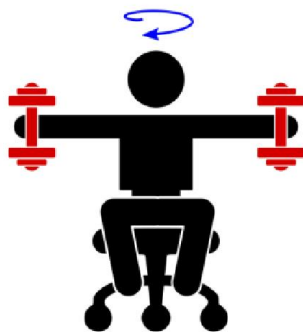
Angular Momentum – $\vec{L}$

$$\vec{L} = I\vec{\omega}$$

$$L = I \cdot \omega$$



$$L = I \cdot \omega$$

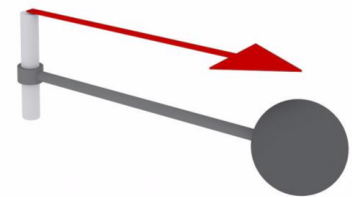




Torque – $\vec{\tau}$

We denote external force $\vec{F}$ exerted on position $\vec{r}$ on the rigid body, therefore

$$\vec{\tau} = \vec{r} \times \vec{F} = \frac{d\vec{L}}{dt}$$



$$\begin{aligned}\vec{\tau} &= \vec{r} \times \vec{F} \\ \vec{L} &= \vec{r} \times \vec{p}\end{aligned}$$



Summary

- Angular Values vs. Linear Values

- Orientation

$\mathbf{R}$

- Angular velocity

$$\vec{\omega} = \frac{\vec{v} \times \vec{r}}{\|\vec{r}\|^2}$$

- Angular acceleration

$$\vec{\alpha} = \frac{d\vec{\omega}}{dt} = \frac{\vec{a} \times \vec{r}}{\|\vec{r}\|^2}$$

- Inertia tensor

$$\mathbf{I} = \mathbf{R} \cdot \mathbf{I}_0 \cdot \mathbf{R}^T$$

- Angular momentum

$$\vec{L} = \mathbf{I} \vec{\omega}$$

- Torque

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

- Position

$\vec{x}$

- Linear velocity

$$\vec{v} = \frac{d\vec{x}}{dt}$$

- Linear acceleration

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{x}}{dt^2}$$

- Mass

$$M = \sum m_i$$

- Linear momentum

$$\vec{p} = M \vec{v}$$

- Force

$$\vec{F} = \frac{d\vec{p}}{dt} = m \vec{a}$$



Application – Billiard Dynamics (1/2)

Even though we have known the elements of rigid body dynamics, the physics in a light billiard game is still complicated...





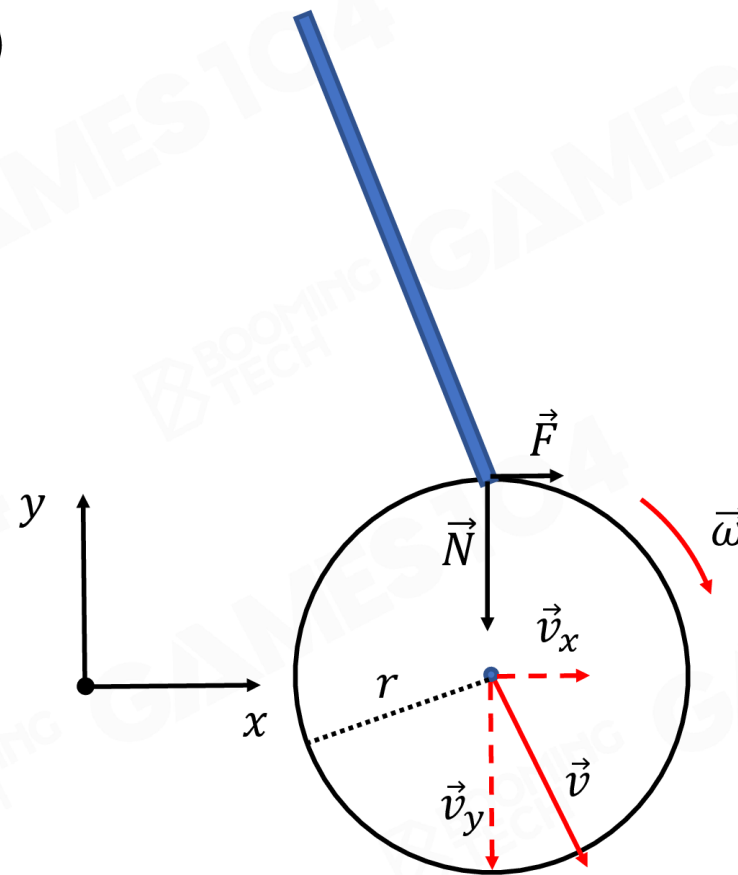
Application – Billiard Dynamics (2/2)

Friction Impulse: $\vec{p}_F = \int \vec{F} dt = m\vec{v}_x$

Pressure Impulse: $\vec{p}_N = \int \vec{N} dt = m\vec{v}_y$

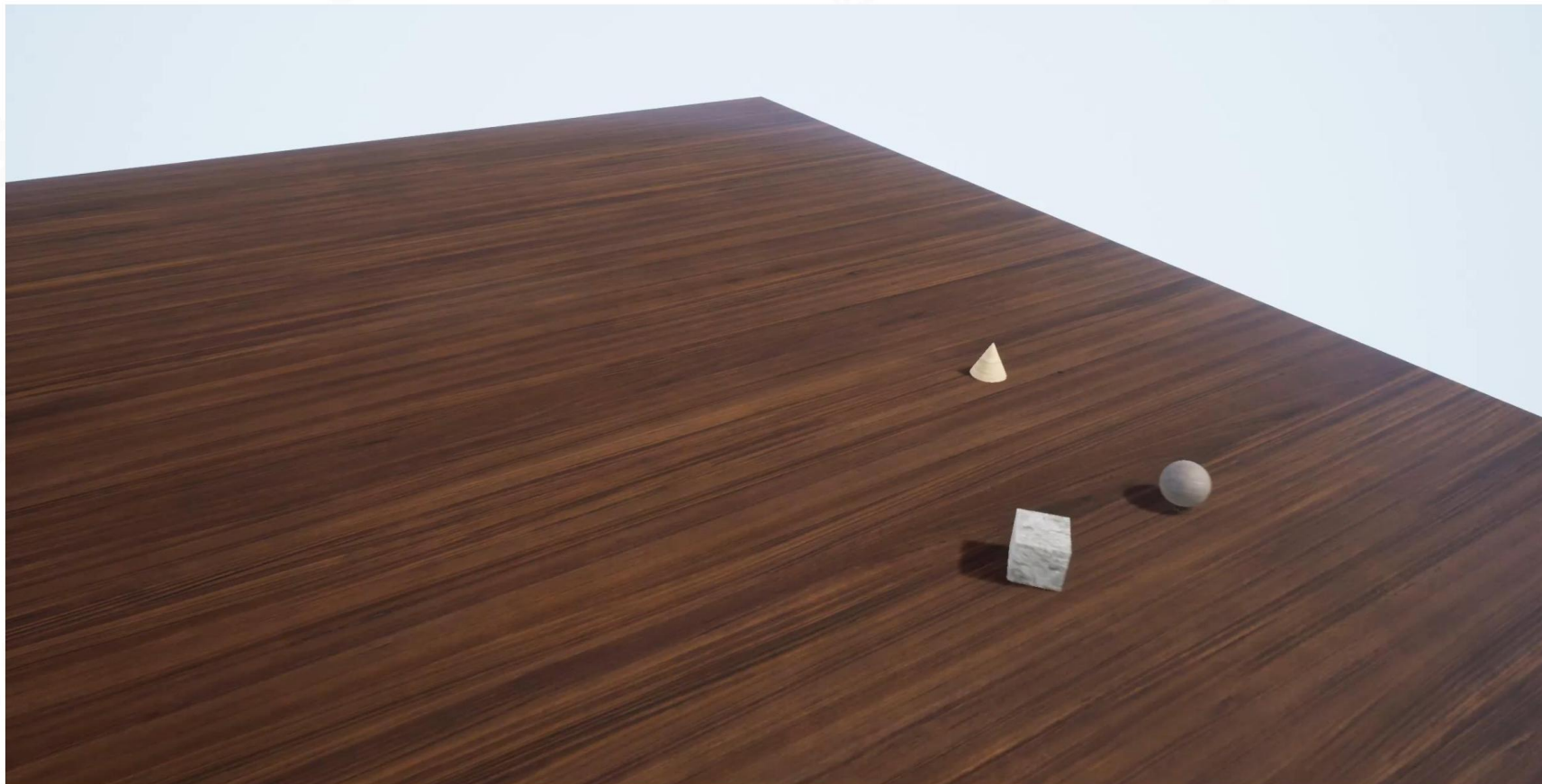
Ball Angular Momentum: $\vec{L}_b = \mathbf{I}_b \vec{\omega} = \vec{p}_F \times \vec{r}_F$

Ball Linear Velocity: $\vec{v} = \vec{v}_x + \vec{v}_y$





Collision Detection



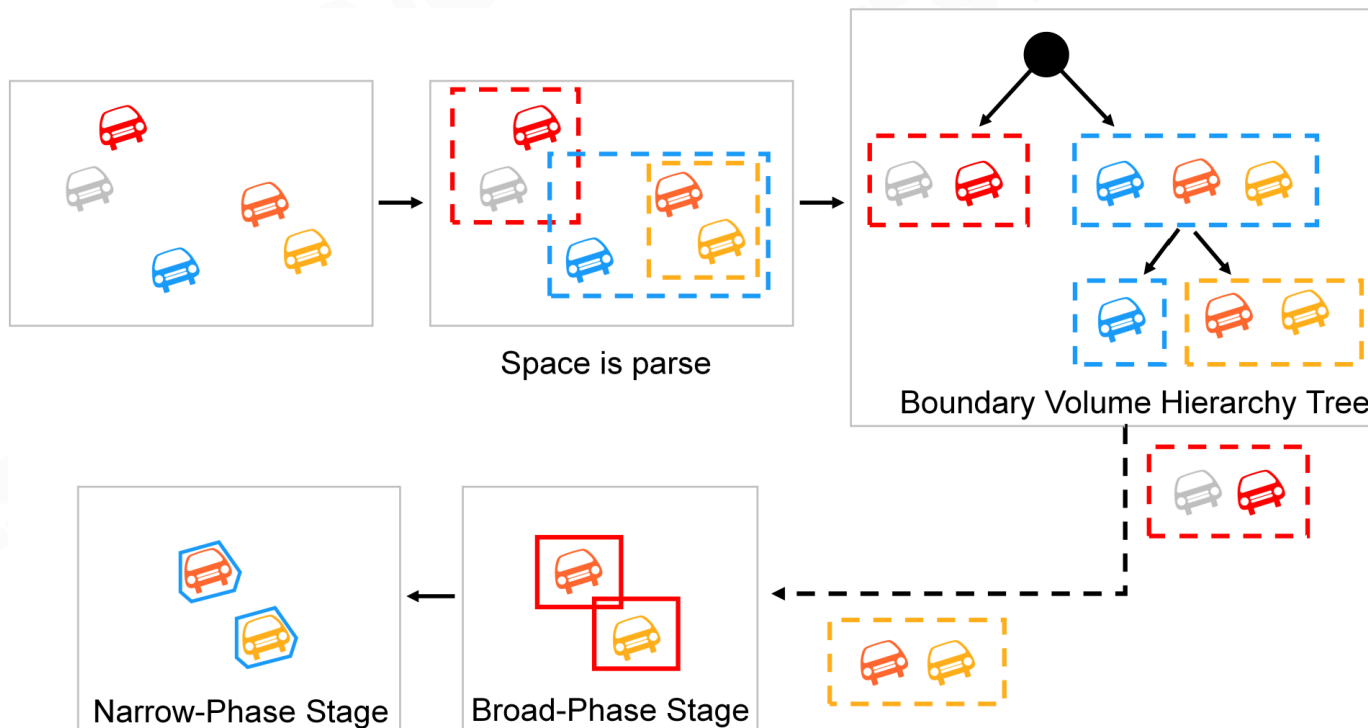


Collision Detection – Two Phases

- Broad phase
 - Find intersected rigid body AABBs
 - Potential overlapped rigid body pairs
- Narrow phase
 - Detect overlapping precisely
 - Generate contact information



Broad Phase and Narrow Phase

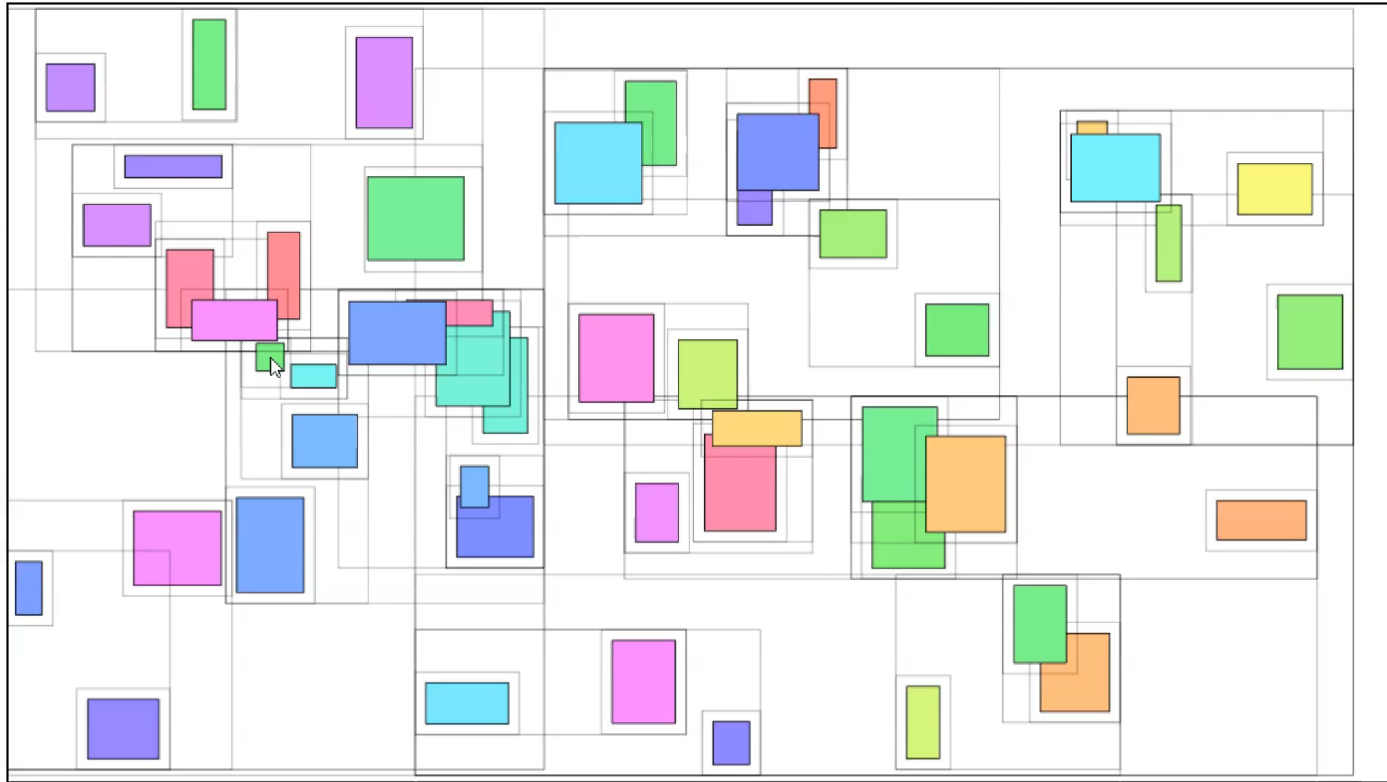




Broad Phase

- Objective
 - Find intersected rigid body AABBs
 - Potential overlapped rigid body pairs
- Two approaches
 - Space partitioning
 - i. e. Boundary Volume Hierarchy (BVH) Tree
 - Sort and Sweep

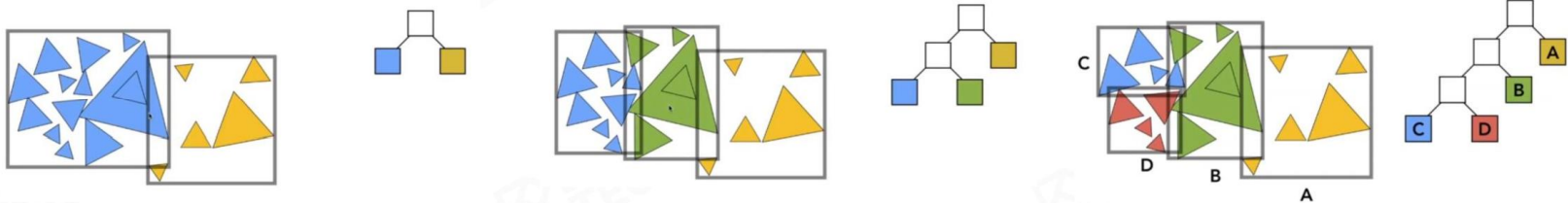
Broad Phase - BVH Tree (1/2)



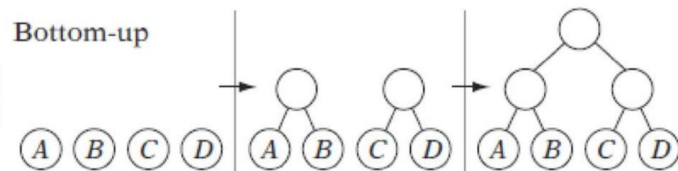
Broad Phase - BVH Tree (2/2)

Recap: Dynamic BVH Tree

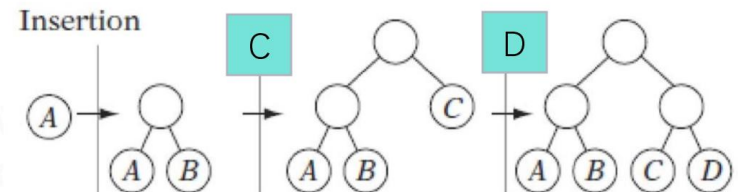
Top-down



Bottom-up



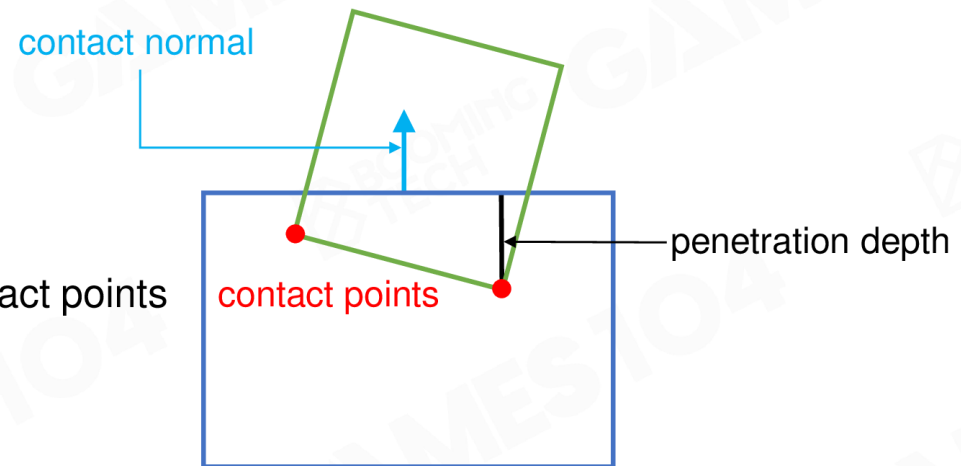
Incremental tree-insertion





Narrow Phase – Objectives

- Detect overlapping precisely
- Generate contact information
 - Contact manifold
 - approximated with a set of contact points
 - Contact normal
 - Penetration depth





Narrow Phase – Approaches

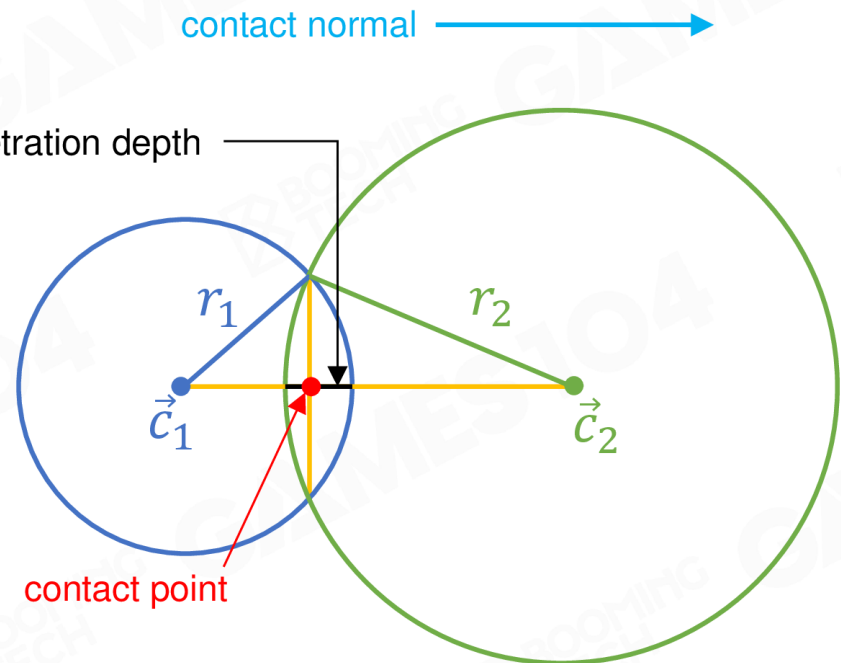
- Three approaches
 - Basic Shape Intersection Test
 - Minkowski Difference-based Methods
 - Separating Axis Theorem



Basic Shape Intersection Test (1/3)

Sphere-Sphere Test

- overlap: $|\vec{c}_2 - \vec{c}_1| - r_1 - r_2 \leq 0$
- contact information:
 - contact normal: $\vec{c}_2 - \vec{c}_1 / |\vec{c}_2 - \vec{c}_1|$
 - penetration depth: $|\vec{c}_2 - \vec{c}_1| - r_1 - r_2$





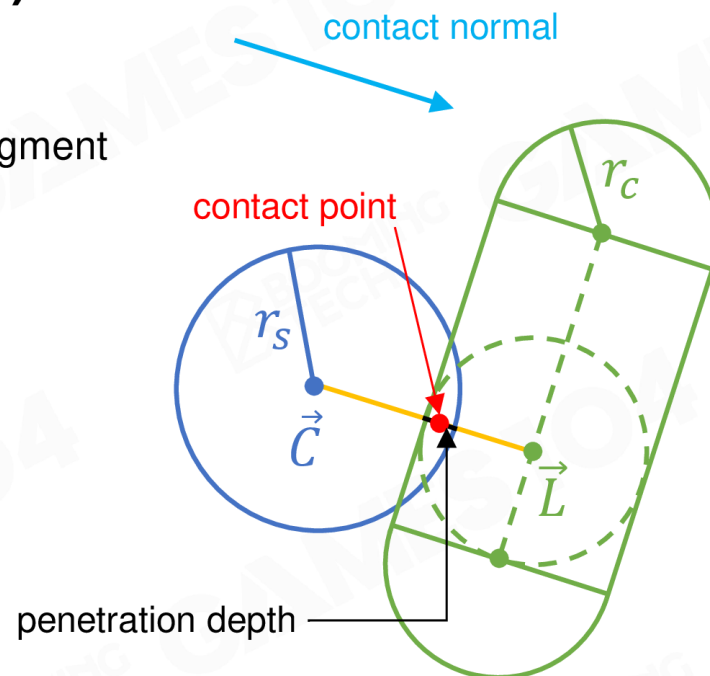
Basic Shape Intersection Test (2/3)

Sphere-Capsule Test

$\vec{L}$ is the closest point on the inner capsule segment

- overlap: $|\vec{C} - \vec{L}| - r_s - r_c \leq 0$
- contact information:
- contact normal: $\vec{L} - \vec{C} / |\vec{L} - \vec{C}|$

penetration depth: $|\vec{C} - \vec{L}| - r_s - r_c$





Basic Shape Intersection Test (3/3)

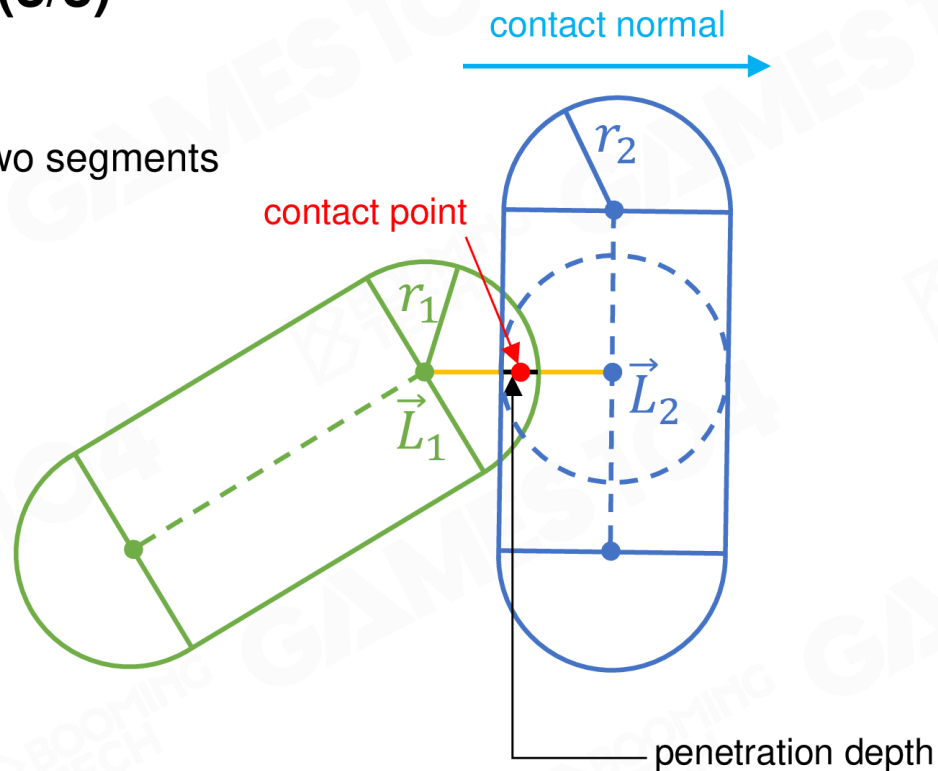
- Capsule-Capsule Test

$\vec{L}_1$ and $\vec{L}_2$ are the closest points on the two segments

overlap: $|\vec{L}_2 - \vec{L}_1| - r_1 - r_2 \leq 0$

contact normal: $\vec{L}_2 - \vec{L}_1 / |\vec{L}_2 - \vec{L}_1|$

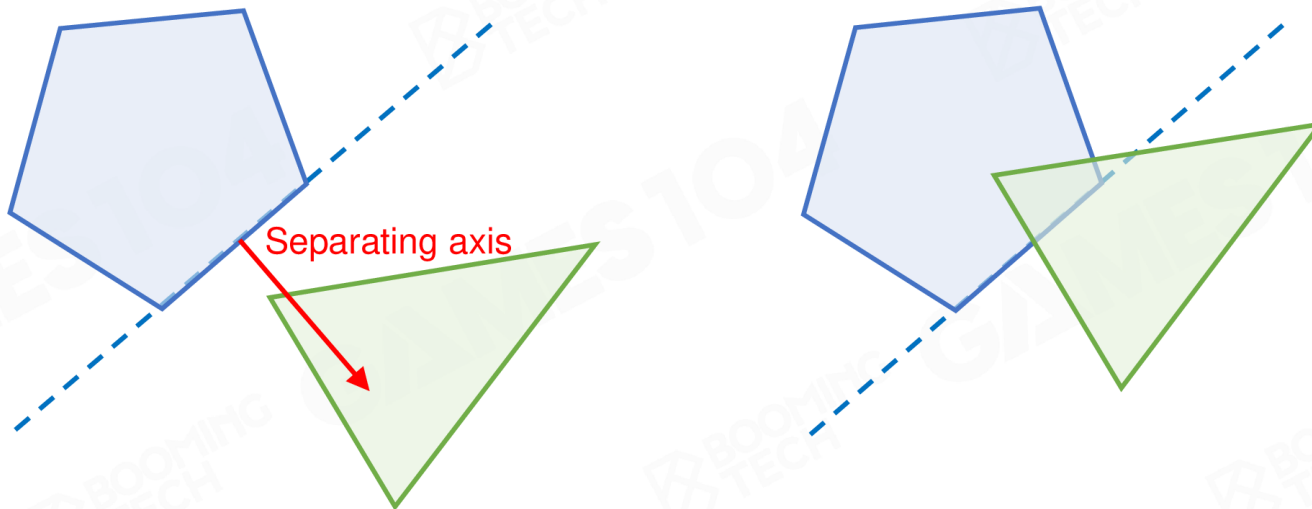
penetration depth: $|\vec{L}_2 - \vec{L}_1| - r_1 - r_2$





Separating Axis Theorem (SAT) - Convexity

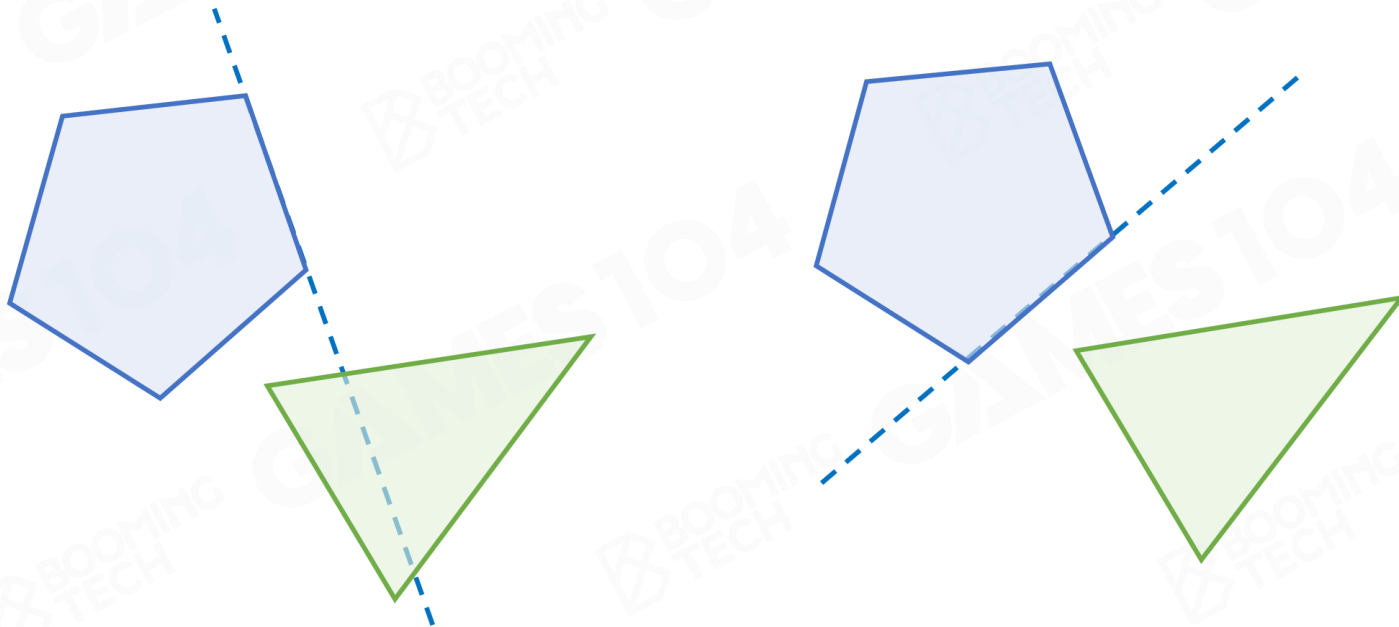
- Edges can separate two convex polygons due to convexity





Separating Axis Theorem (SAT) – Necessity for overlapping

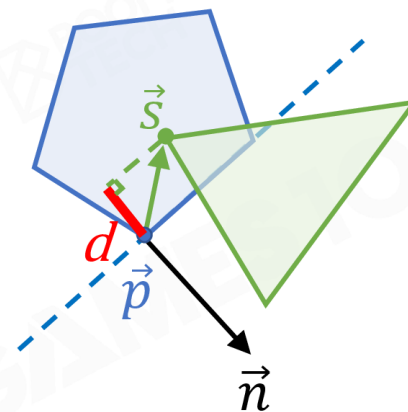
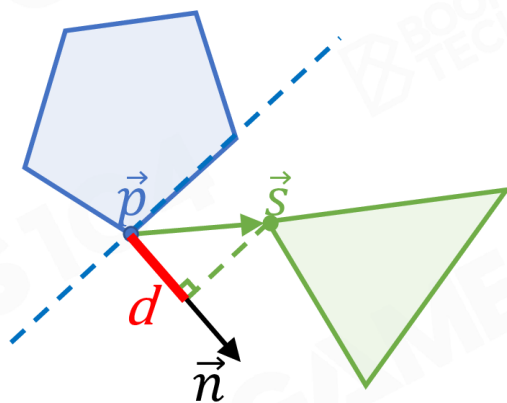
- An edge failed to separate the polygons is not sufficient for overlapping
- All edges must be checked until a separating axis is found





Separating Axis Theorem (SAT) - Separating Criteria

$$d = \vec{n} \cdot (\vec{s} - \vec{p})$$



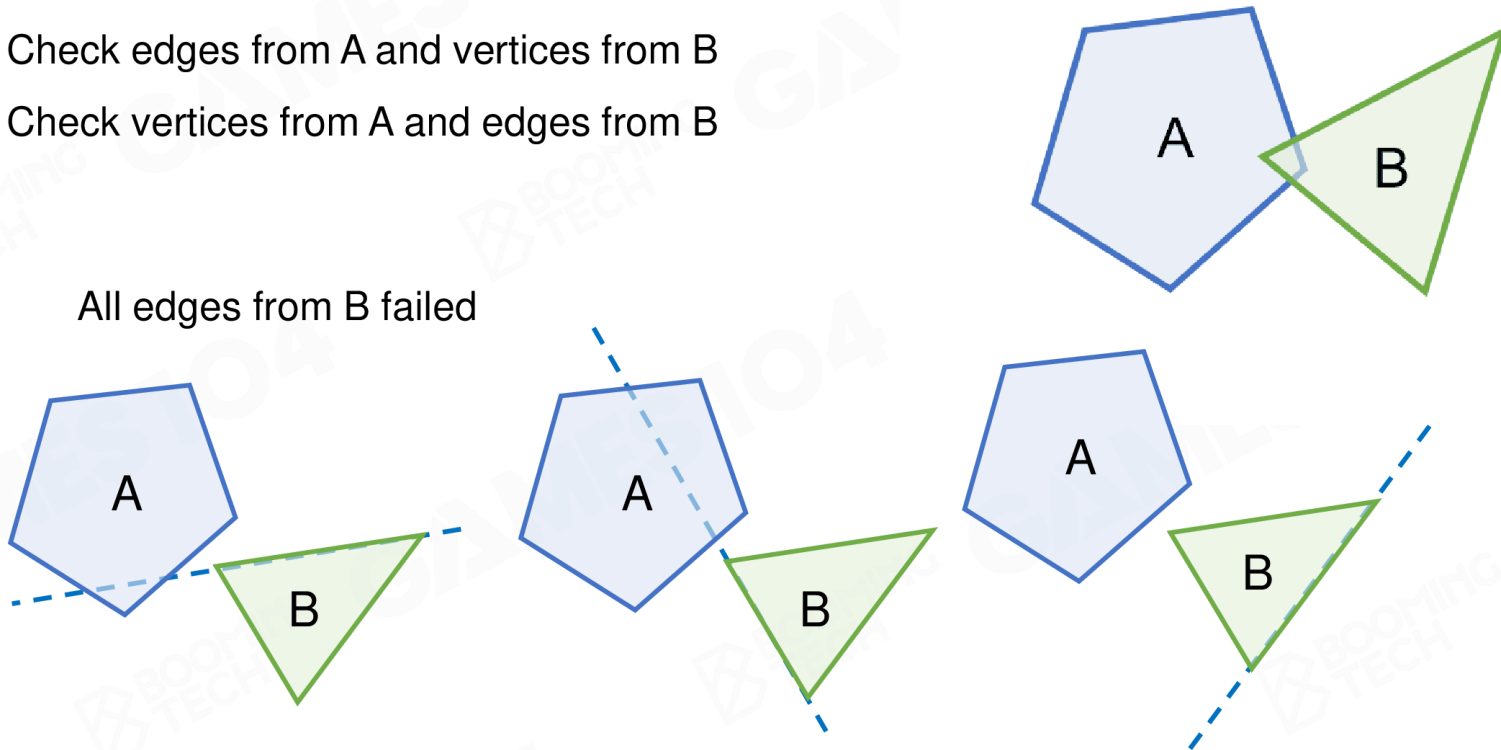
Penetration depth is $|d|$



Separating Axis Theorem (SAT) – 2D Case (1/2)

- Check edges from A and vertices from B
- Check vertices from A and edges from B

All edges from B failed

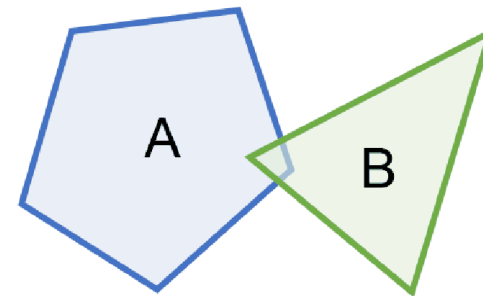




Separating Axis Theorem (SAT) – 2D Case (2/2)

Algorithm 1 SAT-2D

```
1: for each edge  $e_A$  from  $A$  do
2:   overlapped  $\leftarrow$  false
3:   for each vertex  $v_B$  from  $B$  do
4:     if projection of  $v_B$  on normal of  $e_A \leq 0$  then
5:       overlapped  $\leftarrow$  true, break
6:     end if
7:   end for
8:   if not overlapped then
9:      $A$  and  $B$  are separated, terminate
10:  end if
11: end for
12: for each edge  $e_B$  from  $B$  do
13:   overlapped  $\leftarrow$  false
14:   for each vertex  $v_A$  from  $A$  do
15:     if projection of  $v_A$  on normal of  $e_B \leq 0$  then
16:       overlapped  $\leftarrow$  true, break
17:     end if
18:   end for
19:   if not overlapped then
20:      $A$  and  $B$  are separated, terminate
21:   end if
22: end for
23:  $A$  and  $B$  are overlapped, terminate
```





Separating Axis Theorem (SAT) – Optimization for 2D Case

- Check edges from A and vertices from B
- Check vertices from A and edges from B

Optimization

- Cache the last separating axis

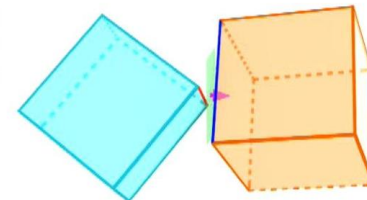
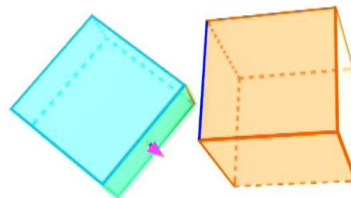
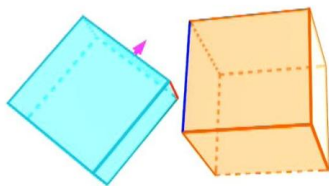
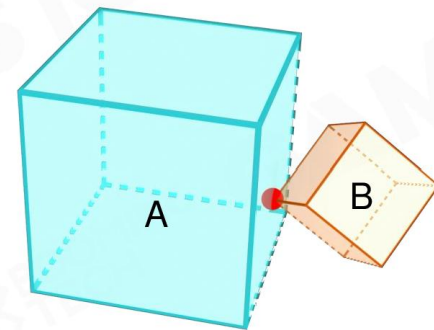
Algorithm 2 SAT-2D-Optimized

```
1: overlapped  $\Leftarrow$  false
2: for each vertex  $v_B$  from  $B$  do
3:   if projection of  $v_B$  on separating_axis_A  $\leq 0$  then
4:     overlapped  $\Leftarrow$  true, break
5:   end if
6: end for
7: if not overlapped then
8:   A and B are separated, terminate
9: end if
10: for each edge  $e_A$  from  $A$  do
11:   overlapped  $\Leftarrow$  false
12:   for each vertex  $v_B$  from  $B$  do
13:     if projection of  $v_B$  on normal of  $e_A$   $\leq 0$  then
14:       overlapped  $\Leftarrow$  true, break
15:     end if
16:   end for
17:   if not overlapped then
18:     update separating_axis_A  $\Leftarrow$  normal of  $e_A$ 
19:     A and B are separated, terminate
20:   end if
21: end for
22: Similar for edges from B
23: ...
```



Separating Axis Theorem (SAT) – 3D Case

- Check faces from A and vertices from B
 - Separating axis: face normals of A
- Check vertices from A and faces from B
 - Separating axis: face normals of B
- Check edges from A and edges from B
 - Separating axis: cross product of two edges





Collision Resolution



Collision Resolution

- We have determined collisions precisely
- We have obtained collision information
- Next, let's deal with collision resolution





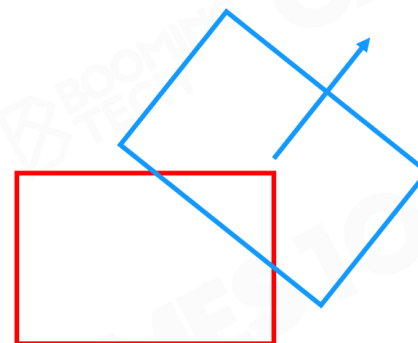
Approaches

- Three approaches
 - Applying Penalty Force
 - Solving Velocity Constraints
 - Solving Position Constraints (will be covered in the next lecture)



Applying Penalty Force

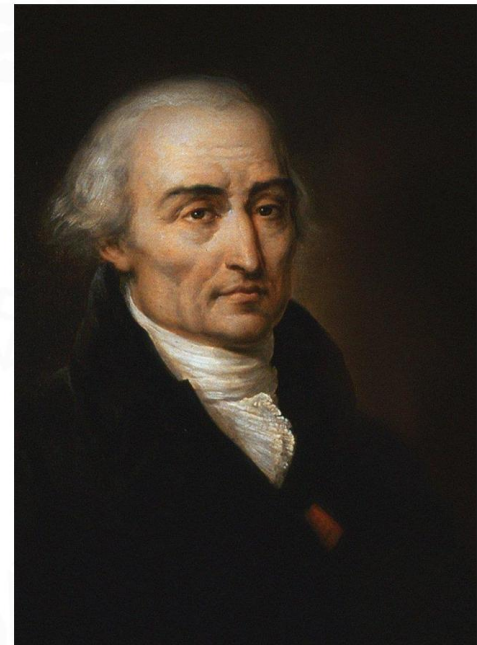
- Rarely used in games
- Large forces and small time steps are needed to make colliding actors look rigid





Solving Constraints (1/2)

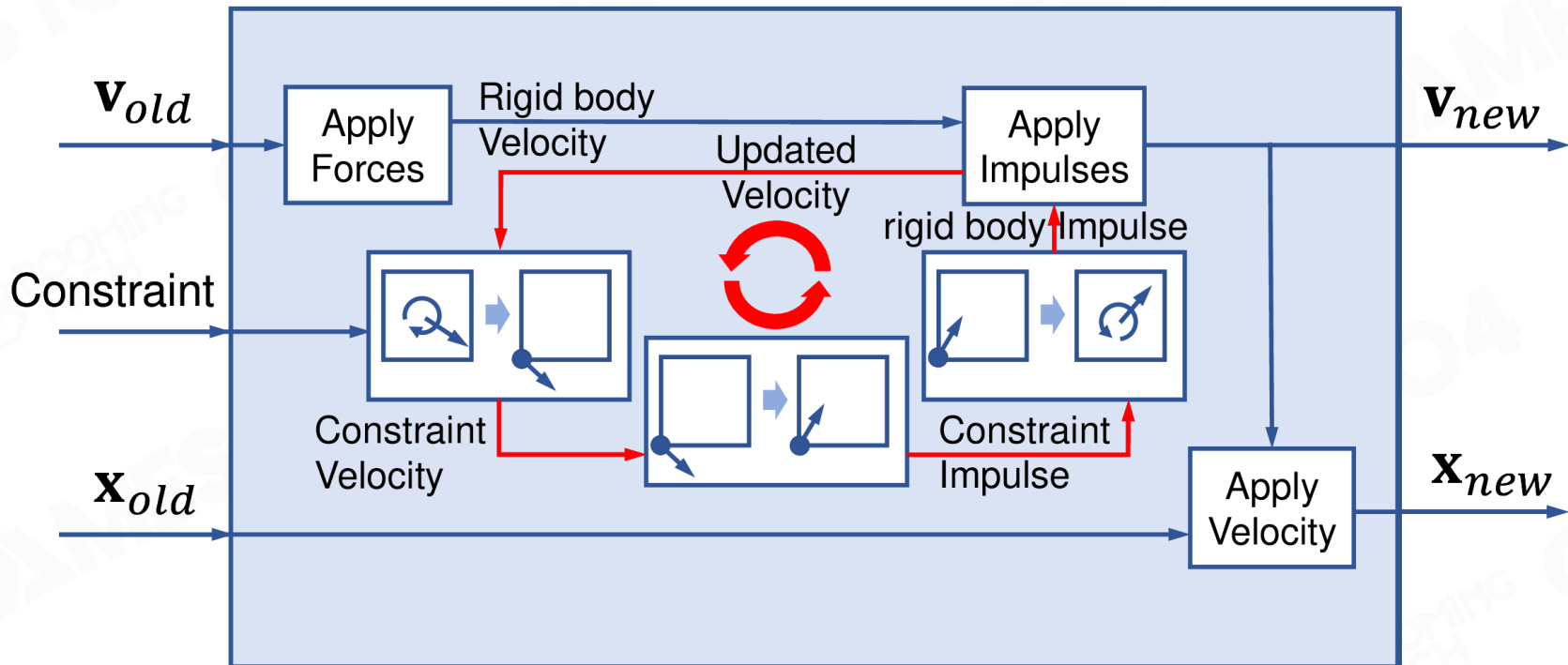
- Modelling constraints based on Lagrangian mechanics
 - Collision constraints
 - Non-penetration
 - Restitution
 - Friction
- Iterative solver



Joseph-Louis Lagrange
(1736 - 1813)



Solving constraints (2/2)





Solving Velocity Constraints

Approaches:

- Sequential impulses
- Semi-implicit integration
- Non-linear Gauss-Seidel Method

Characteristics:

- Fast, stable for most cases
- Commonly used in most physics engines

